

Managing the Rising Cost of Science: Supplier Consolidation in the Academic Life Sciences

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Abstract

Many scientific inputs are sold in increasingly concentrated markets that expose scientists to rising prices, while experimental protocols calibrated to specific products make it difficult to substitute away from them. How much science is lost when these inputs become more expensive, and which scientists are better able to adjust? Using purchase-level procurement records, we follow a merger of two large life sciences suppliers from the prices principal investigators (PIs) pay to the research they produce. Post-merger prices in affected markets rose by roughly 20 percent, while quantities declined only temporarily, indicating limited demand response. PIs facing larger increases in their input costs published less. For the average PI, input costs rose by about 2.8 percent and scientific output fell by about 1.2 percent. The consequences of higher input prices therefore extend beyond research costs to the production of science itself. The output losses fall disproportionately on early-career PIs. Financial resources do not explain this difference. Instead, the ability to adjust appears to improve with experience, consistent with PIs accumulating knowledge of how to respond to higher costs through years of running a lab.

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1 Introduction

Scientific discovery is a resource-intensive process that depends not only on ideas and research talent, but also on experimental inputs purchased from outside suppliers. Many of these inputs are sold in increasingly concentrated supplier markets ([European Commission, 2008](#); [Federal Trade Commission, 2014](#)), exposing scientists to price increases for some of their most essential inputs. Substituting away from inputs whose price has risen is difficult because experimental protocols are calibrated to specific products, making adjustments costly to implement and scientifically risky ([Bradbury and Plückthun, 2015](#); [Freedman et al., 2015](#)). Increases in input prices can therefore translate into higher research costs that scientists cannot easily avoid.

Even so, bearing these higher research costs does not necessarily mean producing less science. Scientists can adjust along many dimensions of the research process, including which experiments to run, whom to hire, and what to purchase from which supplier. Scientists may differ, however, in the advantages they have in responding to higher costs. Financial resources may provide one such advantage, allowing scientists to continue their planned research when it becomes more expensive. Experience may offer another advantage, as scientists accumulate over their careers tacit knowledge about experimental approaches, alternative inputs, and ways of reorganizing projects when research becomes more costly ([Coppola et al., 2025](#)). How scientists respond to higher input costs can therefore reveal not only whether these higher costs affect scientific production, but also what enables some scientists to adjust more effectively than others.

To answer these two questions, we study a rise in scientific input prices arising from supplier consolidation and measure how much scientific output is lost to the higher prices and which principal investigators (PIs) are able to adjust to them. We assemble novel purchase-level procurement records from 165 U.S. public universities spanning 2010 to 2019 and leverage the 2014 merger of Thermo Fisher Scientific and Life Technologies Corporation, two of the largest life sciences input suppliers. The merger increased concentration in markets where the two

firms overlapped in supplying consumables—routinely purchased experimental materials such as reagents, enzymes, and cell culture media—generating plausibly exogenous variation in the prices PIs paid for affected consumables.

We first establish that the merger raised prices in the affected markets, and that the increase translated almost entirely into higher spending. Using a matched difference-in-differences design, we compare product markets with pre-merger competitive overlap against similar, unaffected comparison markets, where a product market is a set of consumables interchangeable at the laboratory bench. Prices in affected markets rose by approximately 20 percent relative to comparison markets. Quantities declined temporarily in both affected and comparison markets, but showed no differential decline in affected markets. Together, these results indicate that demand for these consumables is highly inelastic. Spending therefore increased by roughly the same amount as prices, leaving PIs' research budgets to bear these higher costs.

The merger raises consumables costs differently across PIs because their experimental methods require different bundles of consumables. Our exposure measure captures this variation by weighting each market's estimated price effect by the PI's pre-merger spending share in that market. For PIs linked to procurement records, we construct exposure directly from their purchases. For the remaining PIs, we impute exposure based on the similarity of their publications to those of PIs whose purchases we observe. This approach yields a panel of 13,515 U.S. life sciences PIs, allowing us to compare the research output of more- and less-exposed PIs before and after the merger.

Our analysis finds that these higher consumables costs translate into less scientific output. Event-study estimates show a sustained decline in publications among more-exposed relative to less-exposed PIs following the merger. Since affected markets are only a portion of a PI's consumables purchases, the average PI's consumables costs rise by approximately 2.8 percent, while research output falls by about 1.2 percent. These magnitudes imply an elasticity of scientific output with respect to consumables costs of approximately -0.42 . In dollar terms,

one fewer publication corresponds to approximately \$10,500 in higher consumables costs. The decline in publications therefore suggests that PIs cannot fully adjust to the increase, allowing part of the shock originating in input markets to reach scientific output.

This output loss falls disproportionately on early-career PIs. PIs with fewer than 14 years of experience see output decline as their consumables costs rise, while the response among late-career PIs is smaller and cannot be statistically distinguished from zero. We measure experience as years since a PI's first last-authored publication, our proxy for when a PI begins running their own lab. Experience measured instead from a PI's first publication ever, which adds the years spent at the bench during training, produces a larger and statistically significant difference between early- and late-career PIs.

One potential explanation is that late-career PIs have greater financial resources with which to absorb the cost shock. Yet when we split PIs by proxies for available resources, including baseline funding from the National Institutes of Health (NIH) and the R&D environment of their institutions, we find no comparable gradient. Comparing PIs with similar baseline funding leaves the difference by career stage intact.

We interpret this pattern as consistent with late-career PIs drawing on tacit knowledge accumulated over their careers—knowledge about how to reorganize their research and purchase more efficiently when input costs rise. Preliminary evidence from our survey of currently active life sciences PIs supports the premise that this knowledge is learned by doing rather than transferred across PIs. A large majority of the 19 respondents report that they learned how to manage supply purchasing through trial and error rather than through formal training. We nevertheless view this mechanism as suggestive rather than definitive.

Taken together, the estimated price and output responses illustrate the broader incidence of supplier market power in academic science. Across our sample, a single merger implies approximately \$3.7 million per year in additional spending on consumables alongside roughly 349 fewer papers per year. The research budgets bearing this cost are funded largely by federal

grants from agencies such as the NIH and the National Science Foundation (NSF). In effect, public research funds are redirected toward private suppliers to purchase largely the same consumables at higher prices, while less science is produced.

Our first contribution is to show that the markets supplying scientific inputs can shape the production of science. Existing work studies how the amount and direction of research change with scientists' access to funding ([Jacob and Lefgren, 2011](#); [Li et al., 2017](#); [Azoulay et al., 2021](#); [Howell, 2017](#); [Myers, 2020](#)), research materials, infrastructure, equipment, and technologies ([Murray et al., 2016](#); [Furman and Stern, 2011](#); [Waldinger, 2016](#); [Baruffaldi and Gaessler, 2021](#); [McKeon and Boudou, 2024](#); [Furman and Teodoridis, 2020](#)). Related work shows that lower prices for scientific books increased subsequent research that built on them ([Biasi and Moser, 2021](#)) and that consolidation among academic publishers raised journal prices ([McCabe, 2002](#)). We show that the markets producing and selling the physical inputs used in experiments can directly affect scientific production. When consolidation raises the prices of laboratory consumables, scientific output falls. Because experimental protocols tie PIs to specific products, limited substitution allows market power in upstream input markets to propagate into downstream scientific production.

Our second contribution is to show that experience helps scientists adjust to higher research costs. A large literature documents productivity gains from experience and learning by doing ([Papay and Kraft, 2015](#); [Hendel and Spiegel, 2014](#); [Reagans et al., 2005](#); [Dong et al., 2024](#); [Caplin et al., 2026](#)), while related work shows that different management practices can generate large productivity variation across firms ([Bloom and Van Reenen, 2007](#); [Bloom et al., 2013](#)). We show that experience also matters for how scientists adjust to higher research costs. Output declines are concentrated among early-career PIs, and differences in financial resources do not explain why they experience larger declines. Together with our survey evidence, our findings suggest that experience running a lab may build knowledge that helps scientists reorganize their research when costs rise. Early-career PIs may therefore be particularly vulnerable to cost

shocks because they have had less time to accumulate this knowledge.

Our third contribution is to show the consequences of supplier market power beyond the higher input prices paid by buyers to the production those inputs support. Merger retrospectives typically measure the effects of consolidation through prices paid by buyers ([Ashenfelter et al., 2014](#); [Dafny et al., 2012](#)), while studies of market power in publicly funded markets examine how much public spending is captured by private firms rather than passed through to intended beneficiaries ([Duggan et al., 2016](#); [Cabral et al., 2018](#)). We show that the consequences of supplier market power can extend beyond this transfer when higher input prices reduce the activity those purchases support. Unlike firms producing market goods, PIs generally cannot pass higher input costs on through output prices because most scientific knowledge is not patentable, so higher costs instead reduce the amount of knowledge produced. This reduction is particularly costly because scientists and universities capture only a fraction of the social value generated by their discoveries, so scientific research is already produced below socially optimal levels ([Nelson, 1959](#); [Arrow, 1962](#)). Market power in scientific input markets can therefore compound this distortion by further reducing scientific production.

2 Institutional Setting

Life sciences PIs manage large research labs where they not only oversee the research agenda but also teams of students, postdoctoral fellows, and lab managers. PIs are also responsible for covering the costs of running their lab, including paying the salaries of their research team and purchasing the consumables required for their experiments. These expenses are largely funded by the PI's own research grants, which typically provide a set amount of funding over several years. Thus, PIs must make decisions not only about the research their labs pursue but also about how limited grant funding is allocated across their various research expenses.

In this section, we first explain why PIs have limited ability to substitute away from consumables or expand their research budgets in response to higher prices. We then describe

the supplier industry, where this inelastic demand for consumables coincides with increasing consolidation, and introduce the merger we leverage.

2.1 Managing a Research Lab

PIs facing higher consumables prices cannot easily avoid these increases because both looking for cheaper alternatives and switching products are costly. Finding a lower price for the same product requires comparing prices across suppliers or negotiating discounts from individual sales representatives—a fragmented process that takes time away from research.

Even when cheaper alternatives do exist, PIs face high switching costs. Experimental protocols are calibrated to specific consumables ([Walsh et al., 2007, 2005](#)), so adopting a new supplier's product can require weeks or months of revalidation and carries the risk that experimental results will not reproduce ([Torrington, 2021](#); [Lorsch et al., 2014](#); [Rehman, 2022](#)). Variability in consumables is itself a well-documented source of irreproducibility ([Bradbury and Plückthun, 2015](#); [Weber et al., 2022](#)), and surveys attribute a substantial share of failed replications to the sensitivity of consumables ([Baker, 2016](#); [Freedman et al., 2015](#)). PIs are therefore unlikely to reduce their purchases when consumables become more expensive, making demand for these products likely to be highly inelastic.

PIs also cannot easily increase the total amount of funding available to them to help absorb these higher prices in the short run. Federal grants from agencies such as the NIH and NSF fund the vast majority of academic life sciences research and are the main source PIs rely on for funding personnel, equipment, and supplies ([Azoulay et al., 2019](#)). These grants provide PIs with a set budget to produce research over a period that typically lasts three to five years ([Stephan, 2012](#); [Azoulay et al., 2011](#); [Carnehl et al., 2025](#); [Tham, 2023](#)). Awarded amounts are not typically renegotiated within a grant cycle, so PIs facing higher costs in the middle of their cycle must cover the increase with the funding they already have. PIs are able to seek additional grant funding, but new grants take time to secure and generally arrive in later cycles ([Jacob](#)

and Lefgren, 2011).

Within a grant cycle, most direct costs are also committed to areas a PI cannot easily adjust. Personnel costs represent roughly 80 percent of a typical NIH research budget and cannot be cut on short notice without breaking multi-year commitments (Stephan, 2012; National Institute of Allergy and Infectious Diseases, 2026). Supplies and equipment together account for the remaining 20 percent. A majority of this is spent on consumables because large equipment such as freezers and microscopes is often funded through institutional support and shared across research groups (Ehrenberg et al., 2003; Goldstein and Avasthi, 2021). This makes consumables one of the few margins of spending a PI can adjust within an active grant cycle, yet that flexibility is also constrained, as previously discussed.

A PI facing higher consumables prices is then likely to make adjustments elsewhere in the research process if they must continue purchasing the same consumables and cannot increase their budget in the short run. They may run fewer experiments, defer hiring to later grant cycles, or divert time from research toward raising additional funds. Higher consumables prices can therefore affect research even when the quantity of consumables purchased changes little.

2.2 Market Structure of the Life Sciences Supplier Industry

Consumables are produced and sold by a small number of firms that cater to PIs at universities as well as at biopharmaceutical companies. The inelastic demand described in the previous section becomes particularly consequential as this supplier industry consolidates. When PIs have difficulty substituting away from the consumables they already use, suppliers face little risk of losing buyers after raising prices.

Years of consolidation have left the life sciences supplier industry dominated by a few full-line firms, most notably Thermo Fisher Scientific, Avantor, and MilliporeSigma (Berkrot, 2014; European Commission, 2013). These full-line firms both manufacture their own products and operate broad distribution networks that sell products made by other manufacturers. Smaller

independent manufacturers, such as New England Biolabs, typically specialize in making particular types of consumables and reach university PIs through the distribution networks of these larger full-line firms. In our conversations with procurement offices at several universities, most reported relying heavily on Thermo Fisher Scientific and Avantor, whose life sciences catalogs are often integrated directly into university purchasing platforms.

Universities negotiate with suppliers over prices, but these negotiations provide limited protection against price increases. These contracts typically establish volume discounts from the list prices that suppliers themselves set and can raise. As a result, even a university with a fixed negotiated discount can face higher prices after industry consolidation if a supplier raises its list price. Negotiations are also often decentralized across universities and sometimes across schools or departments within the same university. Because the resulting contracts are private, universities have little information about the prices negotiated elsewhere, so two peer universities can unknowingly pay different prices for the same product ([Innovative Bioscience, n.d.](#); [Donna Kridelbaugh, 2018](#)).

Universities also have limited leverage to discipline higher supplier prices by shifting their purchases elsewhere. They account for less than 15 percent of Thermo Fisher Scientific's revenues, making them a relatively small customer segment ([Thermo Fisher Scientific Inc., 2026](#)). Hospitals and pharmacies, by contrast, can secure lower prices for devices and drugs by steering volume across competing suppliers ([Grennan, 2013](#); [Ellison and Snyder, 2010](#)). Universities have less scope to redirect demand because PIs' experimental protocols often tie their purchases to specific products and suppliers.

2.2.1 The Thermo Fisher Scientific–Life Technologies Merger

One major consolidation in this supplier industry occurred when Thermo Fisher Scientific acquired Life Technologies for \$13.6 billion, combining two of the industry's largest suppliers. At the time of the merger, Thermo Fisher Scientific was the industry's largest full-line supplier

of laboratory instruments and consumables, with \$13.1 billion in annual revenue, while Life Technologies was the second largest supplier of life sciences consumables, with \$3.9 billion in annual revenue ([S&P Global, 2010](#); [Life Technologies Corporation, 2012](#); [Thermo Fisher Scientific Inc., 2014](#)). The deal was announced in April 2013 and completed on February 3, 2014 ([Thermo Fisher Scientific Inc., 2014](#)).

The two firms competed across a broad range of consumables used in molecular and cell biology research. In a subset of these markets, the two firms were among only a few major suppliers. Antitrust authorities expressed particular concern about these markets, which included cell culture media and sera and several gene-modulation products. In some of these markets, the combined firm would hold at least half of worldwide sales and, in others, close to a monopoly ([Federal Trade Commission, 2014](#); [European Commission, 2013](#)).

These concerns led antitrust authorities to require Thermo Fisher Scientific to divest its cell culture media and sera, gene-modulation, and magnetic-bead businesses to the manufacturer GE Healthcare.¹ The sale gave GE Healthcare the assets and intellectual property needed to manufacture these product lines independently. Importantly, however, the divestitures changed who manufactured these products, but not necessarily how or where PIs purchased them. Through its distribution network, Fisher Scientific, Thermo Fisher Scientific sold both its own products and those made by other manufacturers. The divested products could therefore continue to be purchased through Fisher Scientific after GE Healthcare took over their manufacturing process, so the change in manufacturer did not necessarily require PIs to change their purchasing behavior.

¹ See the European Commission's clearance decision in [Case COMP/M.6944](#) and the Federal Trade Commission's analysis of the consent order ([File No. 131-0134](#)).

3 Effect of the Thermo Fisher Scientific–Life Technologies Merger on Prices and Spending

In this section, we show how the merger affected prices and spending in the consumables markets where Thermo Fisher Scientific and Life Technologies competed. We first construct a panel of university consumables purchases and explain how we assign purchases to product markets. We then estimate the merger’s average effects on prices, quantities, and spending. Finally, we show how the price effect varies across consumables, which allows us to capture differences in how the merger affected PIs.

3.1 Data on Consumables Prices

Estimating the merger’s effect on consumables prices requires data on what PIs paid for specific products before and after the merger. We assemble these data from university procurement records, where each line item represents a purchase. Because each university uses a different procurement system, these records are not standardized and contain no common identifier for purchases of the same product. Our main data challenge is therefore to identify the product represented by each purchase and determine which products belong to the same *product market*—defined as a set of consumables that are interchangeable at the laboratory bench.

We address this challenge by building a natural language processing (NLP) classifier that uses item descriptions and supplier names. The classifier works in two steps. It first separates consumables from the non-consumable purchases, and then assigns each consumable purchase to a product market. We train the classifier on a separate set of purchases that we label by hand and then apply it to the procurement records used to estimate the merger’s effects.

3.1.1 Data Sources

We construct two sets of data. The *price data* contain the purchases used to estimate the merger's effects on prices, quantities, and spending. The *training data* contain labeled examples used to train the NLP classifier. Table 1 summarizes the sources used to construct each dataset, which we describe in turn below. Appendix Figure A6 provides an overview of how these sources combine to form the two datasets and where each enters our analysis.

Table 1. Data Sources

Source	Obtained from	Coverage
Panel A: Price data		
GovSpend procurement extract	Contract with commercial procurement-data vendor	4.8 million purchases (2010–2019) 165 public universities
Oregon State University records	FOIA request	49,087 purchases (2010–2019) All departments
Panel B: Training data		
<i>Labeled by product market and consumable status</i>		
UT Dallas procurement records	FOIA request	61,882 purchases (2011–2024) Departments of Biological Sciences and Chemistry & Biochemistry
UMich procurement records	FOIA request	306,883 purchases (2010–2019) Departments of Molecular, Cellular and Developmental Biology, Biological Chemistry, and Pharmacology
<i>Labeled as non-consumables only</i>		
California state procurement records	California Open Data Portal	162,717 purchases (2012–2015) Non-life sciences state purchase orders
Thermo Fisher Scientific negotiated price list	2025 New York State contract	5,997 purchases Furniture, equipment, and apparel categories

Notes: FOIA denotes the Freedom of Information Act. A purchase is a single line item in a procurement record. Panel A lists the sources of the price data, from which we estimate the merger's effects. Panel B lists the sources of the training data, the labeled purchases on which we train the classifier. The UT Dallas and UMich purchases are assigned to product markets and the resulting markets are labeled as consumable or non-consumable. The California records and price list contribute non-consumable examples only.

Price Data. GovSpend is a commercial procurement-data vendor that aggregates publicly available purchase records from U.S. public entities through open records requests and partnerships

with institutional purchasing portals. We obtain an extract of 4.8 million life sciences purchases across 165 public universities from 2010 to 2019. Each purchase reports key information including the item description, price paid, quantity ordered, total spending, purchase date, supplier name, and purchasing university. Appendix B describes how we construct this extract.

We also obtain additional procurement records through Freedom of Information Act (FOIA) requests filed with public universities. Nine universities provided records with purchase-level prices from before the merger: East Carolina University (ECU), Texas Tech University (TTU), University of Kansas (UKansas), University of Florida (UF), University of South Florida (USF), University of Texas at Austin (UT Austin), University of Texas at Dallas (UT Dallas), University of Michigan (UMich), and Oregon State University (OSU). Eight of these universities are already included in the GovSpend extract, so we only add the procurement records from OSU to the price data.

Training Data. We construct the training data from four sources. Procurement records from UT Dallas and UMich provide purchases that we assign by hand to product markets and also label as consumable or non-consumable. California state procurement records and Thermo Fisher Scientific’s negotiated price list with the State of New York provide additional item description examples of non-consumable purchases.

The UT Dallas and UMich records offer different advantages for constructing the training data. The UT Dallas records report supplier catalog numbers (SKUs) in addition to item descriptions and supplier names, allowing us to identify products that cannot be reliably identified from the description and supplier alone. While the UMich records do not include SKUs, UMich is the largest university in our FOIA data and its records span a broader set of life sciences departments, which expands the range of products represented in the training data.

Because the first classification step must distinguish consumables from a broad range of non-consumable purchases, we add two sources of non-consumable training examples. First, we use publicly available California state procurement records from 2012–2015 and keep 162,717

purchases whose United Nations Standard Products and Services Code (UNSPSC) falls in segments unrelated to life sciences consumables.² Second, we use Thermo Fisher Scientific's 2025 negotiated price list with the State of New York and keep 5,997 products listed under furniture, equipment, and apparel categories.³ From both sources, we use only the item descriptions as training data.

3.1.2 Classification of Purchases into Product Markets

The classifier operates in two steps: (1) determining whether a purchase is a consumable and (2) assigning consumable purchases to product markets. We first explain how we define product markets and construct the training labels. We then describe the classification algorithm.

Product-Market Definitions and Labeled Training Data. To identify which products are interchangeable at the laboratory bench, we define product markets using Thermo Fisher Scientific's online product catalog ([Fisher Scientific, 2026](#)). As the largest full-line supplier and distributor of life sciences products ([Federal Trade Commission, 2014](#)), Thermo Fisher Scientific offers a catalog that covers products from a wide range of manufacturers, making it a useful reference for identifying products purchased from many different suppliers.

We use the catalog to assign each unique item description–supplier pair in the UT Dallas and UMich records to a product market. We search for the purchased product using the item description and supplier name and, for UT Dallas purchases, also use SKUs when available. Once we identify the corresponding product, we record the “product type” listed on its catalog page and use this as the initial product-market assignment. See Appendix Figure [A7](#) for an example.

² The kept categories include paper products (14); machinery, vehicles, tools, and components (21–27, 30–32, 39–40); information technology, office, printing, safety, cleaning, service-industry, and recreational equipment and supplies (43–49); food, appliances, apparel, timepieces, published products, furniture, and musical and educational products (50, 52–60); and services and real property (70–95). The excluded categories, 10–13, 15, 20, 41, 42, 51, and 85, cover live plants and animals, raw materials, chemicals, resins and polymers, fuels, mining equipment, laboratory and testing equipment, medical supplies, pharmaceuticals, and healthcare services.

³ Apparel/Personal Protective Supplies, Basic Equipment, Appliances, Furniture, and Equipment Peripherals and Supplies.

For products covered by the European Commission’s (EC) analysis of the merger, we refine these initial assignments using the EC’s findings on substitutability ([European Commission, 2013](#)). In some cases, the EC distinguishes products within a catalog product type that are not interchangeable in practice. For example, Thermo Fisher Scientific lists fetal bovine serum (FBS) as a single product type, whereas the EC finds that FBS from different geographic origins is not substitutable. We therefore separate FBS into product markets based on country of origin.

We then hand-label each product market as consumable or non-consumable. Consumables markets include products such as *pipette tips*, *latex gloves*, and *Taq DNA polymerase*, whereas non-consumable markets include products such as *desktops*, *lab cabinets*, and *hotplates*. To train the classifier’s first step, which distinguishes consumable from non-consumable purchases, we combine these UT Dallas and UMich labels with the additional non-consumable examples from the California procurement records and Thermo Fisher Scientific price list described previously. The resulting training data contain 368,765 unique item description–supplier pairs assigned to 2,091 product markets, of which 1,105 are consumable markets.

Classification Algorithm. The first step of the classifier separates consumables from non-consumables. We combine keyword rules with a logistic regression on term frequency–inverse document frequency (TF-IDF) representations of the item descriptions and supplier names.⁴ The keyword rules classify clear cases directly. For instance, item descriptions containing keywords such as *polymerase* or *antibody* are classified as consumables, and descriptions containing keywords such as *shipping* or *furniture* are classified as non-consumables. The logistic regression uses the training data and classifies the remaining purchases.

The second step assigns each consumable purchase to one of the 1,105 product markets using nearest-neighbor matching in TF-IDF space. To distinguish closely related markets with similar descriptions, we use keyword rules for terms that clearly identify a particular market. For example, descriptions containing “taq” and “polymerase” but not “hot start” are assigned

⁴We ran a BERT-based alternative and found that the TF-IDF text representations outperformed it at substantially lower computational cost.

to *Taq DNA Polymerase* rather than *Hot Start DNA Polymerase*. When a keyword rule and the nearest-neighbor match disagree, the keyword rule takes precedence.

Some product markets that we can distinguish by hand have item descriptions that are too similar for the classifier to distinguish reliably. In these cases, we combine them into a broader market for classification and estimation. For example, primary antibodies targeting different antigens are not substitutes and are therefore labeled as separate markets by hand. However, their procurement descriptions are too similar for the classifier to reliably assign purchases to the correct product market, so we combine them into a single *Primary Antibodies* market.⁵

3.1.3 Final Estimation Sample

We impose three restrictions on the classified purchases to construct the final estimation sample. First, to keep the composition of universities stable over time, we retain universities that report consistently from 2010 to 2019.⁶

Second, we keep only product markets that the classifier assigns reliably. We evaluate our classifier on an 80/20 train/validation split of the UT Dallas and UMich purchases. For each product market, precision is the share of purchases assigned to that market that belong to it, and recall is the share of purchases that belong to that market that are correctly assigned to it. Both scores fall when there are few labeled training examples for that product market. Thus, we only keep product markets with at least 25 labeled examples and out-of-sample precision and recall of at least 0.8. We refer to product markets that meet both thresholds as high-confidence markets. These markets account for 55 percent of consumables spending in our price data (\$114 million). Appendix Table A2 reports average precision and recall scores for both steps of the classifier.

Finally, we exclude 34 markets that experienced other major shocks during the sample

⁵ We aggregate markets that relate to labware, such as beakers, cuvettes, and vials, and markets such as antibodies and cell lines. These markets lie outside the two firms' competitive overlap, so this aggregation has little scope to influence our estimated merger effects.

⁶ A university is defined as reporting consistently if its purchase records cover at least eight months of each year.

period, so that price movements unrelated to the merger do not enter our estimates. These include laboratory filtration following Danaher's 2015 acquisition of Pall and synthetic DNA oligonucleotides following the entry of Twist Bioscience and the expansion of Integrated DNA Technologies. None of these excluded markets is among the markets in which Thermo Fisher Scientific and Life Technologies competed before the merger.

The final estimation sample contains 514,660 purchases from 73 universities across 152 product markets, accounting for 36 percent of total consumables spending in our price data (\$75.5 million). We aggregate these purchases to the university–product market–year level for estimation. Appendix Table [A1](#) provides descriptive statistics on the top universities, suppliers, and product markets in the final estimation sample.

3.1.4 Price, Quantity, and Spending Variables

Our records report the amount paid and number of packages purchased, but not the size of each package. We therefore do not observe the price per unit of the product. For example, we can see how many bottles of fetal bovine serum a PI purchased and how much they paid, but not whether each bottle contained 50 mL or 500 mL.

Because we do not observe package size, within each university–product market–year we measure price as the average amount paid per purchase rather than the product's unit price, and quantity as the number of purchases rather than the amount of product purchased. Spending is the product of these two measures. In our estimation, we use all three outcomes in logs. We log quantity and spending after aggregating to the university–product market–year. For price, we instead log the amount paid for each purchase before averaging across purchases. Table [2](#) summarizes these definitions.

A shift in package size would also raise our price measure. For example, if products were sold in larger packages after the merger, the amount paid per purchase would increase even if the unit price remained unchanged. Larger packages would also reduce the number of purchases

needed to buy the same amount of product. We therefore examine price and quantity together to assess whether changes in our price measure reflect genuine changes in unit price or changes in package size.

Table 2. Key Variables

Variable	Symbol	Measure
Price	P_{jmt}	Average amount paid per purchase by university j in product market m and year t
Quantity	Q_{jmt}	Number of purchases by university j in product market m and year t
Spending	S_{jmt}	Total amount paid by university j in product market m and year t

Notes: A purchase is one line item in a procurement record. In the estimation, quantity and spending are logged after aggregation to the university–product market–year. For price, the amount paid for each purchase is logged before averaging across purchases. Dollar amounts are measured in current dollars.

3.2 Average Merger Effects on Prices, Quantities, and Spending

We first establish that the merger raised prices on average across treated product markets, with little relative change in quantities and a corresponding increase in spending. To estimate the overall effect, we pool treated markets and their matched controls and estimate an event-study specification.

Defining Treated and Control Markets. The EC identified twelve consumables areas in which Thermo Fisher Scientific and Life Technologies had pre-merger competitive overlap ([Case COMP/M.6944](#)). Because our product markets are narrower than these areas, we map our high-confidence product markets into the EC’s analysis. We define a product market as treated ($D_m^T = 1$) if it falls within one of these twelve areas or is complementary to a product market that does.⁷ Of our 152 high-confidence product markets, 46 are treated under this definition, including 15 complementary markets. Although not directly mentioned by the

⁷ A natural alternative would be to compute the Herfindahl–Hirschman Index (HHI) directly from purchasing records, since HHI captures pre-merger market concentration. Implementing this in our data is complicated because GovSpend records supplier names as free-text fields rather than standardized identifiers, so the same firm often appears under multiple variants (e.g., “Thermo Fisher Scientific,” “Thermo Fisher Sci.,” and “ThermoFisher”). While manual cleaning of these fields helps, residual noise still propagates into any firm-level market share calculation and makes HHI a noisy proxy for market concentration.

EC, these complementary products are purchased and used alongside products that are—for example, running buffers required for pre-cast gels—so their prices may also respond to the merger. Our estimates are robust to excluding these markets. Appendix C maps our treated product markets to the EC’s analysis.

All remaining high-confidence markets in the final estimation sample of Section 3.1.3 serve as candidate controls ($D_m^T = 0$). For each treated market m^* , we construct a set of up to two matched controls from these candidate control markets using nearest-neighbor matching on the pre-period average and slope of log prices. Matching is performed with replacement, so the same control market may be matched to multiple treated markets. This procedure yields 51 distinct control markets across the 46 treated-market comparisons.⁸

3.2.1 Empirical Specification

We pool the 46 treated markets and their matched controls in a single event-study regression for each outcome,

$$y_{jmt} = \sum_{\substack{n=-4 \\ n \neq -1}}^5 \beta_n \mathbf{1}\{t - 2014 = n\} \times D_m^T + \gamma_j + \delta_m + \lambda_t + \epsilon_{jmt}, \quad (1)$$

where y_{jmt} is the log outcome (price, quantity, or spending) in market m for products purchased by university j in year t , $\mathbf{1}\{t - 2014 = n\}$ indicates that year t is n years from the merger year 2014, D_m^T equals one for treated markets, and γ_j , δ_m , and λ_t are university, market, and year fixed effects. The specification omits $n = -1$, so each β_n gives the difference between treated and matched control relative to the year immediately before the merger. Observations are weighted by each market’s total 2013 spending, and standard errors are clustered at the market level.

Identification rests on the common-trends assumption that, absent the merger, treated and matched control markets would have followed the same counterfactual trajectory in each

⁸ Appendix Figures A1, A2, and A3 reproduce the three event studies without matching, comparing treated markets to the full set of candidate control markets.

outcome. We examine pre-merger trends to assess the validity of this assumption.

3.2.2 Results

We find that treated product markets experienced persistent and growing price increases following the merger. Quantities temporarily fell after the merger, but by similar amounts in treated and control markets. As a result, higher prices in treated markets translated into higher spending.

Figure 1 presents the price result in two panels: Panel (a) plots the pooled difference between treated and control markets, and Panel (b) plots the treated and control price trajectories separately, each normalized to the year before the merger. The flat pre-trend in the pooled difference supports the plausibility of the identifying assumption behind equation (1).⁹ Following the merger, the pooled difference jumps in 2014 and climbs steadily, from roughly 11 percent in the first post-merger year to 23 percent by 2019, averaging roughly 20 percent over the post-merger period. The separate trajectories by treatment status show that this divergence is driven by rising prices in treated markets, while prices in control markets remain relatively flat aside from a slight decline from the second post-merger year onward.

Treated quantities do not fall relative to controls after the merger, consistent with the idea that substituting away from these products is difficult. Figure 2 plots the pooled difference between treated and control quantities in Panel (a) and their separate trajectories in Panel (b). Despite the substantial relative price increase in treated markets, the pooled quantity difference in Panel (a) remains statistically indistinguishable from zero in every year, which suggests highly inelastic demand. Panel (b) shows a temporary decline of about 10 percent in both treated and control markets in the year immediately after the merger. This common decline may reflect a temporary tightening of research budgets that reduced consumables purchases across all markets.

⁹ We also find no evidence of anticipatory responses. Appendix Table A4 reports a placebo test that assigns a false 2012 treatment date to the pre-merger panel.

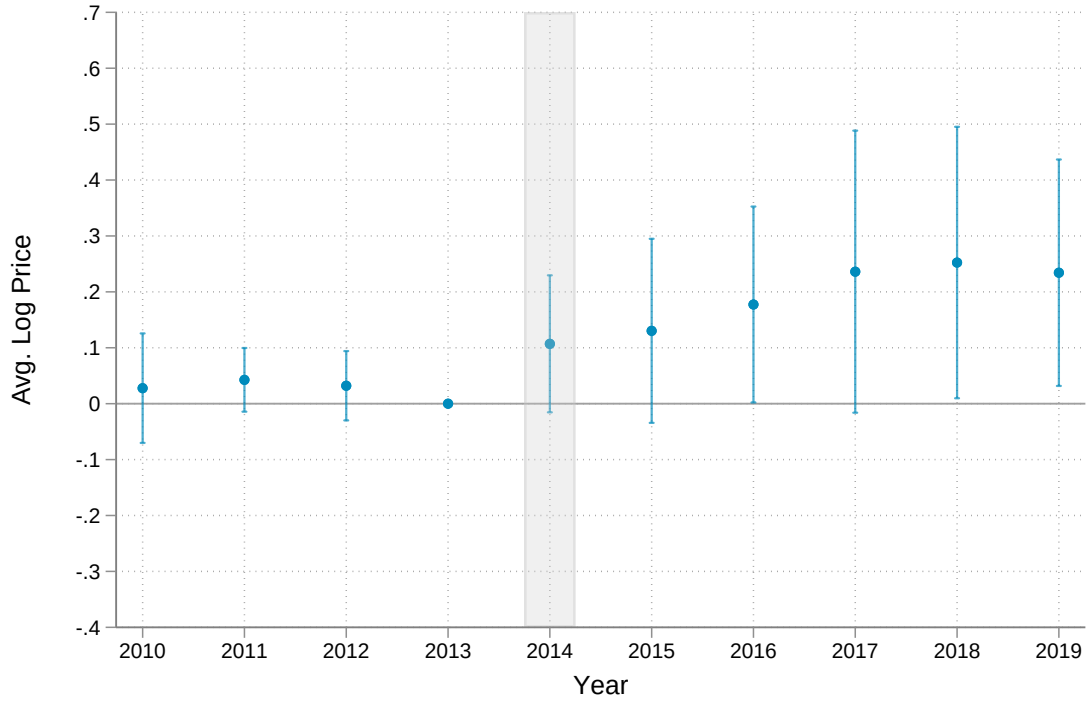
Because quantities did not differentially change between treated and control markets, the relative increase in treated prices translates into higher treated spending. Panel (a) of Figure 3 shows a pooled spending increase of roughly 30 percent. Panel (b) shows that treated spending rises steadily after the merger, while control spending remains relatively flat aside from a small decline.

Together, the price, quantity, and spending effects characterize demand for the treated consumables as highly inelastic. Treated quantities barely move relative to controls despite the roughly 20 percent price increase, so the higher price passes almost entirely into higher spending on the same consumables rather than into substitution toward cheaper alternatives or lower purchase frequency. Appendix Table A3 reports pooled difference-in-differences estimates of these effects and their robustness across alternative fixed-effects specifications.

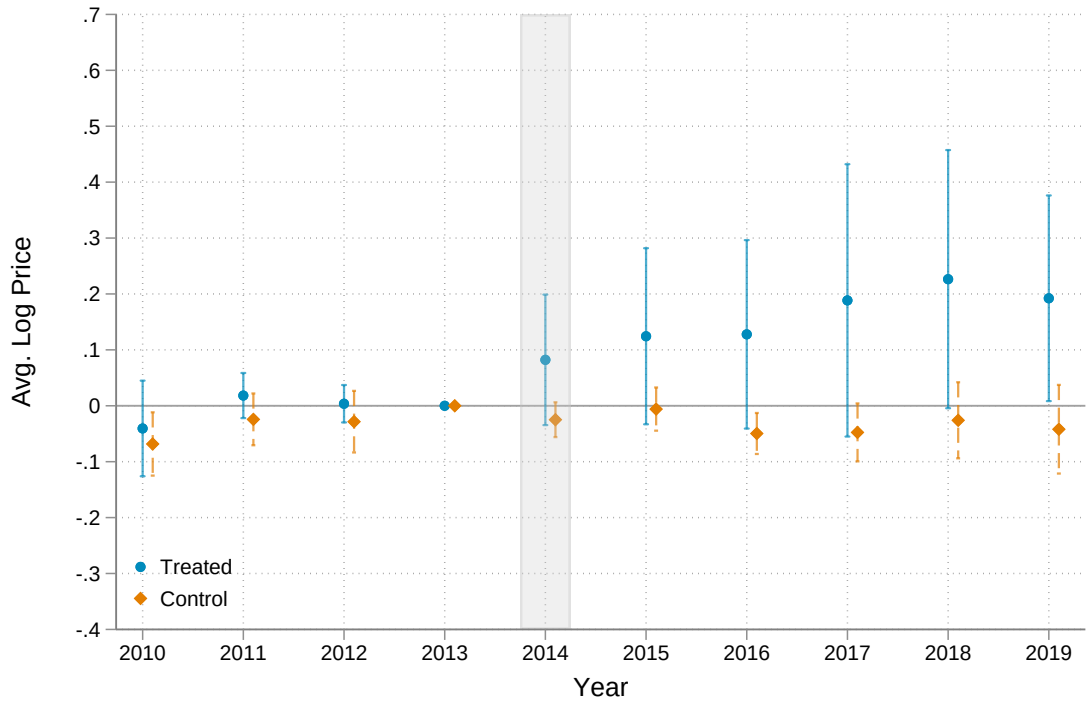
The absence of a differential quantity response also weighs against a shift in package size as the source of the price increase. Since price is measured per purchase, a shift toward larger packages in treated markets would raise measured price while lowering the number of treated purchases relative to controls. Treated and control quantities instead move together throughout the sample. If anything, the number of treated purchases rises slightly after 2015, though not significantly. This would be consistent with a shift toward smaller packages, which would bias our measured price increase downward rather than upward. The price increase therefore likely reflects a change in the unit price of the same set of products rather than a change in package size.

Figure 1. Pooled Merger-Induced Price Effects

(a) Pooled difference



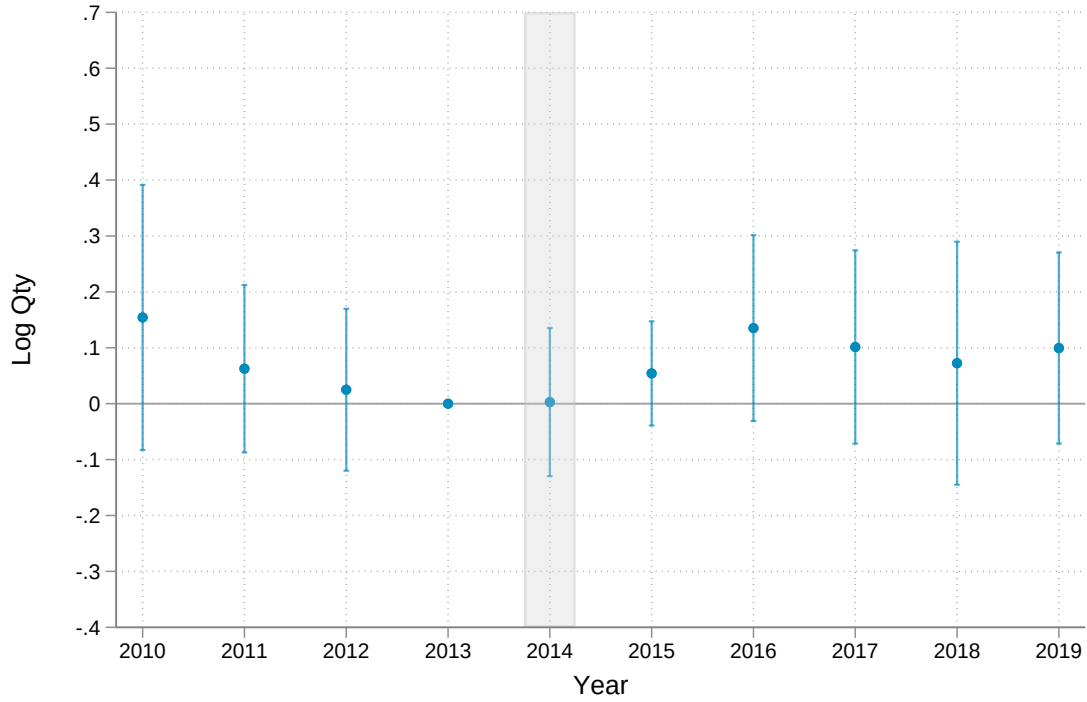
(b) Treated and control market effects



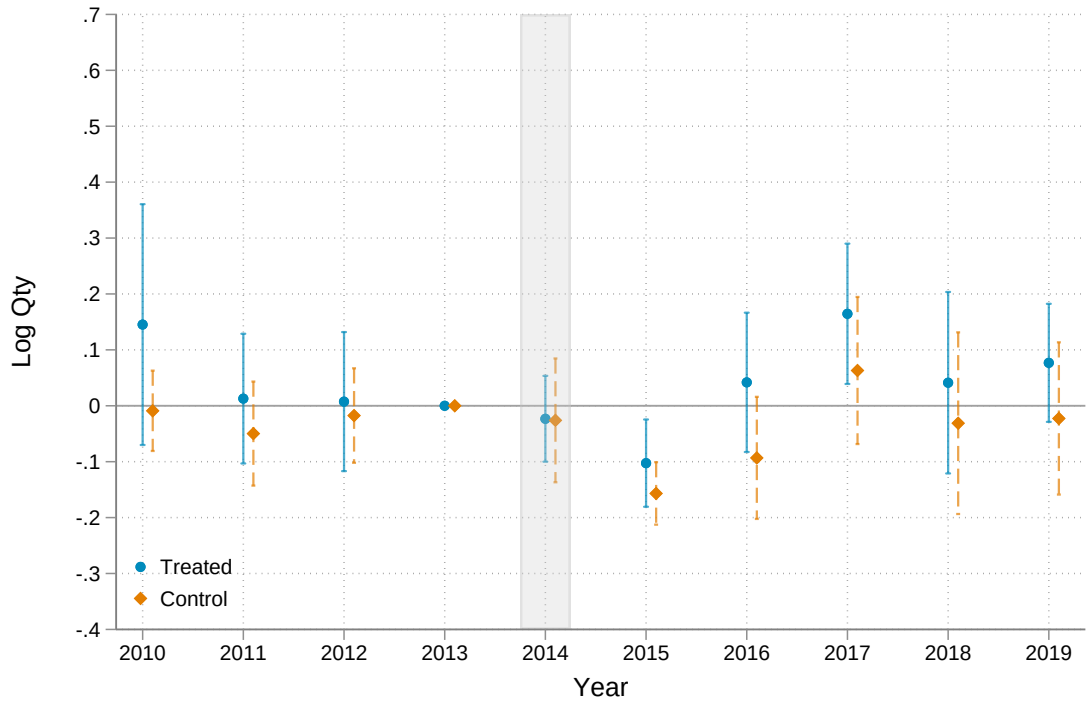
Notes: Event-study estimates of average log prices around the Thermo Fisher Scientific–Life Technologies merger. Panel (a) plots the estimated coefficients $\hat{\beta}_n$ from equation (1), the difference between treated product markets and their matched controls n years from the merger, normalized to zero at $t = -1$. Panel (b) plots the estimated year effects $\hat{\lambda}_t$ as the control market effects (orange) and $\hat{\lambda}_t + \hat{\beta}_n$ as the treated market effects (blue), both linearly detrended using the control market path, so the vertical gap between the two in each year equals the Panel (a) coefficient. The vertical gray band indicates the merger year (2014). The average price in $t = -1$ is \$143.30 in treated markets and \$104.07 in matched control markets.

Figure 2. Pooled Merger-Induced Quantity Effects

(a) Pooled difference



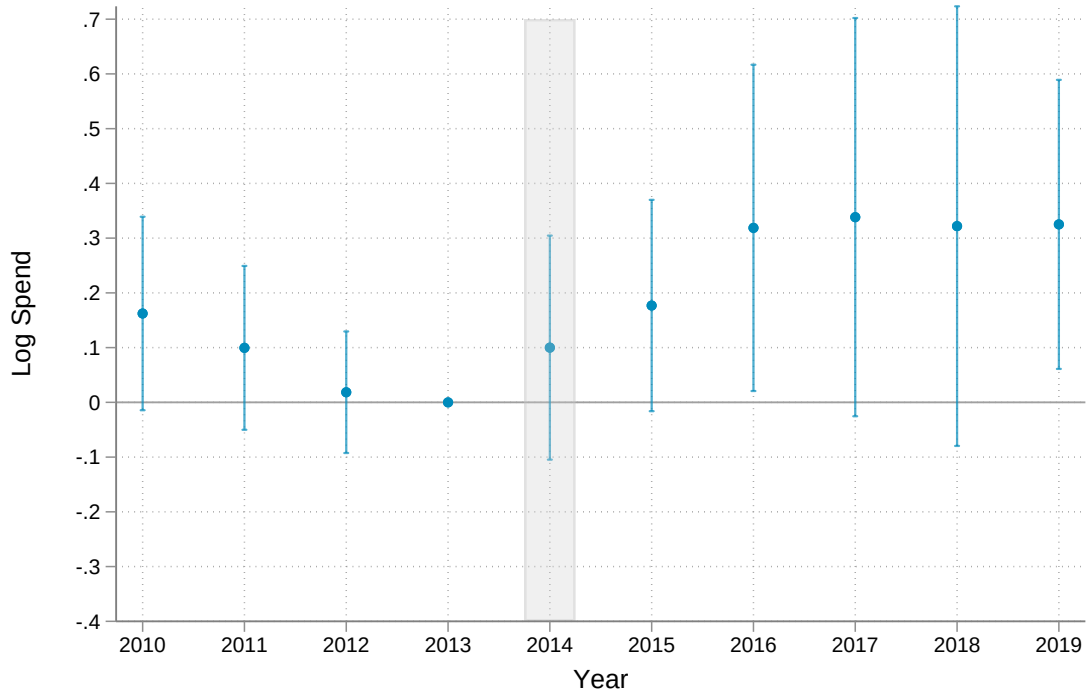
(b) Treated and control market effects



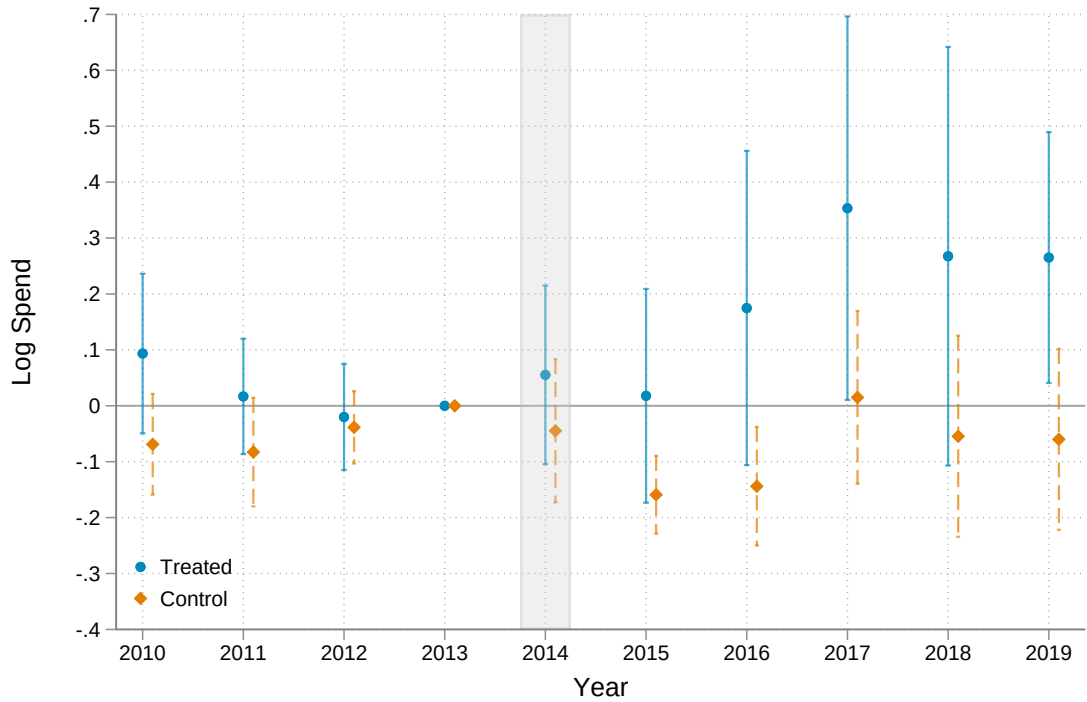
Notes: Event-study estimates of average log quantity around the merger. Panel (a) plots the estimated coefficients $\hat{\beta}_n$ from equation (1), the difference between treated product markets and their matched controls n years from the merger, normalized to zero at $t = -1$. Panel (b) plots the estimated year effects $\hat{\lambda}_t$ as the control market effects (orange) and $\hat{\lambda}_t + \hat{\beta}_n$ as the treated market effects (blue), both linearly detrended using the control market path, so the vertical gap between the two in each year equals the Panel (a) coefficient. The vertical gray band indicates the merger year (2014). The average quantity in $t = -1$ is 17.8 purchases in treated markets and 25.7 in matched control markets.

Figure 3. Pooled Merger-Induced Spending Effects

(a) Pooled difference



(b) Treated and control market effects



Notes: Event-study estimates of average log spending around the merger. Panel (a) plots the estimated coefficients $\hat{\beta}_n$ from equation (1), the difference between treated product markets and their matched controls n years from the merger, normalized to zero at $t = -1$. Panel (b) plots the estimated year effects $\hat{\lambda}_t$ as the control market effects (orange) and $\hat{\lambda}_t + \hat{\beta}_n$ as the treated market effects (blue), both linearly detrended using the control market path, so the vertical gap between the two in each year equals the Panel (a) coefficient. The vertical gray band indicates the merger year (2014). Average spending in $t = -1$ is \$2,094 in treated markets and \$1,983 in matched control markets.

3.3 Market-Level Price Effects

We next show that the price increases underlying the average effect vary substantially across treated markets. We estimate a separate price effect for each treated market using a difference-in-differences design with the same matched control markets as before. This market-level variation matters because PIs differ in the composition of their consumables bundles, so the same merger can affect PIs differently.

3.3.1 Empirical Specification

For each treated market m^* , our difference-in-differences specification replaces the relative-year indicators of equation (1) with a post-merger indicator $\text{Post}_t = \mathbf{1}\{t \geq 2014\}$ and is estimated using that treated market and its matched controls:

$$y_{jmt} = \beta_{m^*}^P \cdot D_m^T \cdot \text{Post}_t + \gamma_j + \delta_m + \lambda_t + \epsilon_{jmt}, \quad (2)$$

where y_{jmt} is the average log price in market m for products purchased by university j in year t , and $\beta_{m^*}^P$ is the estimated merger price effect for treated market m^* . The terms γ_j , δ_m , and λ_t are university, market, and year fixed effects. Observations are weighted by 2013 market spending, and standard errors are clustered at the market level.

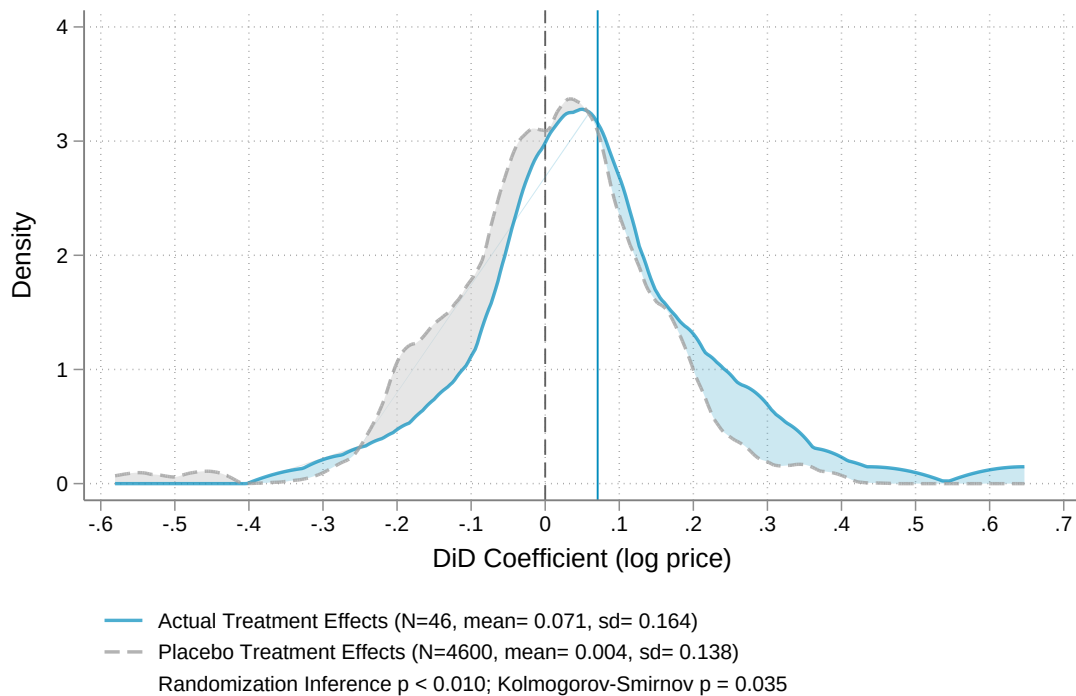
Estimating equation (2) separately for each treated market produces the 46 market-specific price effects, $\{\hat{\beta}_m^P\}_{m=1}^{46}$. Because each estimate is based on a relatively small sample of university–product market–year observations, these estimates are more likely to be noisy. We therefore apply empirical Bayes shrinkage to reduce this noise and use the shrunk estimates throughout the remainder of the paper. Appendix D details the procedure.

3.3.2 Results

Figure 4 shows the distribution of estimated price effects across treated markets and compares it with a placebo distribution constructed by randomly assigning treatment status across control markets. The placebo distribution captures the dispersion in market-level estimates that can arise from sampling variation alone.

The treated distribution is centered well above zero, at 0.071, whereas the placebo distribution is centered at zero. This unweighted mean of the market-level effects is smaller than the pooled estimate from Section 3.2.2 because the pooled specification gives greater weight to higher-spending treated markets. Randomization inference rejects the null that the mean treated effect could be generated by placebo treatment assignments and a Kolmogorov–Smirnov test rejects equality of the treated and placebo distributions.

Figure 4. Market-Level Merger-Induced Price Effects



Notes: The figure plots the unweighted distribution of the market-specific difference-in-differences price effects $\hat{\beta}_m^p$ across the 46 treated markets (solid) against a placebo distribution (dashed). The placebo distribution pools 100 random reassignments of treated status among the control markets, each the size of the treated set, for 4,600 placebo estimates. The same empirical Bayes shrinkage is applied to both distributions. Randomization inference compares the mean treated effect to the 100 placebo means ($p < 0.01$), and a Kolmogorov–Smirnov test compares the two distributions ($p = 0.035$).

4 The Effect of Higher Input Prices on Scientific Research Output

This section examines how the higher consumables prices generated by the merger affect scientific research output. We measure research output using an annual panel of PI publications and measure each PI's exposure to higher prices based on the increase in the cost of their pre-merger consumables bundle. We first describe the data and construction of these measures. We then estimate the effect of higher consumables costs on research output and examine the mechanisms behind our results by asking which PIs are better able to adjust to the shock.

4.1 Data Linking PI Spending to Publications

Estimating the effect on research output requires measuring both a PI's publications and the increase in their consumables costs generated by the merger. We measure research output using annual publication data, while measuring the cost increase is more difficult because the procurement data from Section 3 do not identify who at the university made each purchase.

We therefore combine publication records from two bibliographic databases, PubMed and OpenAlex, with procurement records obtained through FOIA requests from five universities that identify the PI or federal award associated with each purchase. The publication records allow us to track PI research output and characterize the type of research they do, while the FOIA records provide pre-merger spending across product markets at the PI level. We combine these data with the market-level price effects estimated in Section 3 to construct PI-level exposure to the merger-induced price increases. Appendix Figure A6 summarizes how the data from this and the previous section combine to form the exposure measure and estimation sample.

4.1.1 Publication Records

We measure each PI's research output using annual publication counts from two large bibliographic databases, PubMed and OpenAlex. We use PubMed to define the universe of life sciences publications and OpenAlex to link these publications to individual PIs across years. The records also provide titles, abstracts, and MeSH terms, which we use to measure similarity in the type of research conducted by PIs.

PubMed Medical Subject Headings (MeSH) Extract. PubMed is the National Library of Medicine's database of biomedical and life sciences publications. We use its Medical Subject Headings (MeSH), a curated taxonomy of biomedical topics, to define our universe of life sciences publications. Our query covers the MeSH descriptors *Anatomy, Chemicals and Drugs, Organisms, Phenomena and Processes, Diseases, Anesthesia and Analgesia, Therapeutics, and Clinical Study*, which jointly capture basic, translational, and clinical life sciences research.¹⁰ We restrict to journal articles and exclude comments, case reports, technical notes, letters, and reviews.

Our measure of research output is the number of journal articles on which a given PI is the *last author*, consistent with authorship norms in the life sciences. The first author on a life sciences paper is typically the scientist who did most of the experimental work, usually a graduate student or postdoctoral fellow, while the last author is typically the PI, who funds and oversees the research.

OpenAlex. OpenAlex is a comprehensive open catalog of the global research system developed by the nonprofit OurResearch (Priem et al., 2022). Its key advantage over PubMed is that it provides consistent author and institutional identifiers that allow us to easily track PI publication records over time. OpenAlex aggregates and standardizes information from sources including PubMed, Microsoft Academic Graph, Crossref, and ORCID. It disambiguates author names

¹⁰ Our results are robust to dropping the clinical life sciences papers from our sample (not reported).

using names, publication records, citation patterns, and ORCID identifiers, and standardizes institutional affiliations using a machine learning model.

Authors often list multiple affiliations, so we assign each PI a single home institution per year from their full publication history. We use the affiliation they list most consistently and, when multiple affiliations are listed equally often, use the affiliation from surrounding years.¹¹

4.1.2 PI Spending

Among the FOIA records introduced in Section 3.1.1, purchases from five universities can also be linked to individual PIs: UT Dallas, UMich, UT Austin, TTU, and ECU. In addition to prices, these records report either the name of the PI responsible for the purchase or the federal award number used to fund it. When only an award number is available, we identify the PI associated with the award using NIH RePORTER, NSF Award Search, USAspending.gov, and publication acknowledgments. With this approach, we are able to attribute 402,515 of the 479,423 purchases (87% of spending) to a PI.

As described in Section 3.1.2, the UT Dallas and UMich purchases have hand-assigned product-market labels, while purchases from the other three universities are assigned to product markets using our classifier. Together, the PI links and product-market assignments allow us to measure each PI's pre-merger spending by product market.

4.1.3 PI Exposure to Higher Input Prices

We summarize each PI's exposure to the merger as a shift-share of the market-level price effects, weighting each market's price increase by the PI's pre-merger spending share in that market. We observe this measure directly for PIs whose purchases we can identify and impute it for the remaining PIs using publication text similarity.

¹¹ We keep separately branded hospitals distinct from their affiliated universities, so that Mass General Brigham, for example, is not merged into Harvard. When an author lists two affiliations in the same year, we assign the one that was primary in the previous year, or otherwise the one primary in the following year.

Construction. A PI's exposure measures the percentage increase in the cost of their pre-merger consumables bundle. We construct it by weighting each market's estimated price effect by the PI's pre-merger spending share in that market:

$$\text{Exposure}_i = \sum_m s_{im} \hat{\beta}_m^P \quad (3)$$

where s_{im} denotes PI i 's pre-merger spending share in product market m and $\hat{\beta}_m^P$ is the empirical Bayes shrunk merger-induced price effect in market m from equation (2). Intuitively, a PI who spent half of their pre-merger consumables budget in unaffected markets and half in a market where the merger increased prices by 20 percent has an exposure measure of 0.1, corresponding to a 10 percent increase in the cost of their pre-merger bundle.

We calculate s_{im} relative to the PI's total pre-merger spending across the high-confidence markets defined in Section 3.1.3. Because $\hat{\beta}_m^P$ is estimated only for treated markets, only spending shares in treated markets enter the exposure measure. These shares therefore sum to the fraction of the PI's high-confidence pre-merger spending in treated markets, rather than to one. We denote this fraction by $S_i = \sum_{m \in \mathcal{T}} s_{im}$, where $\mathcal{T}$ is the set of treated markets, and refer to it as the PI's treated-market share.

The exposure measure varies across PIs for two reasons: (1) differences in the composition of their pre-merger consumables spending (the s_{im} shares), which reflect different research focuses and experimental protocols, and (2) differences across markets in the merger-induced price effects (the $\hat{\beta}_m^P$ shifts). PIs who spent more in markets with larger merger-induced price increases therefore have higher exposure. Some PIs have a negative exposure measure, reflecting a decrease in the cost of their pre-merger bundle. We keep these PIs in our sample because dropping them would condition on the treatment variable and discard identifying variation.¹²

Imputation. The exposure measure can be calculated directly only for PIs whose pre-merger

¹² Appendix Table A6 reports the estimate for the positive-exposure subsample. The estimate has the same sign but is less precise, as excluding PIs with a negative exposure measure removes identifying variation.

purchases we observe, and these PIs can serve in the imputation only if they have enough pre-merger publications to draw on. Among the PIs identified in the FOIA records, 208 meet both requirements, and we refer to them as FOIA PIs.

For each non-FOIA PI whose publications we observe, we impute the exposure measure based on the similarity of their pre-merger research to that of the FOIA PIs. The underlying assumption here is that PIs working on similar scientific topics are likely to use similar experimental protocols and therefore have comparable spending compositions across consumables markets. We measure research similarity using the cosine similarity between TF-IDF representations of each PI’s pre-merger publication text, constructed from titles, abstracts, and MeSH terms. A PI’s imputed exposure measure is then a similarity-weighted average of the observed exposure measures of their three most textually similar FOIA PIs:

$$\widehat{\text{Exposure}}_i = \sum_{k \in \mathcal{N}_3(i)} w_{ik} \cdot \text{Exposure}_k, \quad (4)$$

where $\mathcal{N}_3(i)$ denotes the set of three nearest FOIA PIs to target PI i and w_{ik} are similarity weights normalized to sum to one. We choose three neighbors by validating the imputation on coauthors of FOIA PIs.¹³

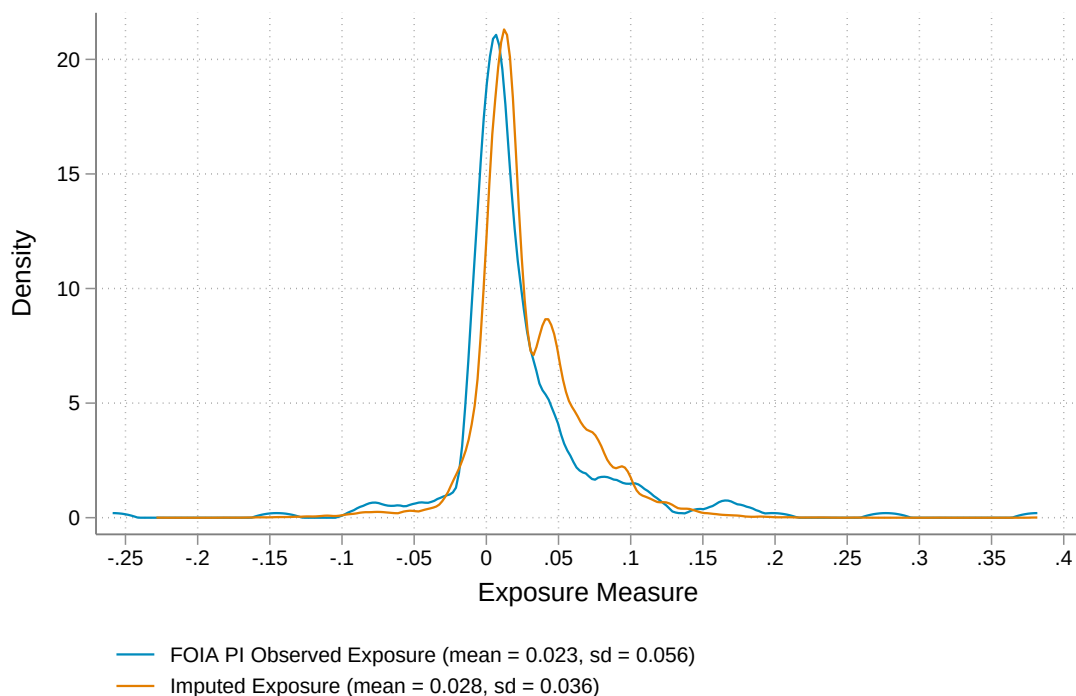
To ensure that the exposure measure is imputed only for PIs working in research areas represented among the FOIA PIs, we group PIs into thirty research clusters using k-means clustering of pre-merger publication text. We use the imputed exposure measure only for PIs whose cluster contains at least one FOIA PI.

The imputed exposure measures are centered close to the observed ones but are less dispersed, since averaging over three neighbors dampens dispersion. Figure 5 compares the distribution of the imputed exposure measure for non-FOIA PIs with the observed distribution

¹³ We use the FOIA coauthor’s observed exposure measure as the benchmark for a PI whose own purchases we do not observe. For each such PI, we predict the exposure measure from their independent publication text, excluding any papers written jointly with the FOIA PI, and compare the prediction to the benchmark. Prediction accuracy peaks at three neighbors and declines on either side, which motivates the choice of $\mathcal{N}_3(i)$.

among FOIA PIs.¹⁴

Figure 5. Observed and Imputed PI Exposure Distributions



Notes: The blue line shows the distribution of the directly observed exposure measure among the 208 FOIA PIs. The orange line shows the distribution of the imputed exposure measure for the 13,433 non-FOIA PIs in the panel.

4.1.4 Estimation Sample

We construct the estimation sample from U.S. life sciences authors who appear as the last author on at least one journal article between 2010 and 2019. We restrict to PIs at R1 or R2 institutions under the *Carnegie Classification* and to PIs who do not move institutions during our time period.¹⁵ We also require that each PI's research cluster contain at least one FOIA PI.

We construct a balanced annual panel of PIs from 2010 to 2019, recording publication counts as zero in years that a PI does not publish. The final panel comprises 13,515 PIs at 198 universities, with an average of 2.16 publications per PI-year before the merger. Of the 208 FOIA PIs used for the imputation, 82 also meet the sample criteria and enter the panel, and we

¹⁴ The treated-market share S_i is imputed with the same neighbors, and Appendix Figure A4 compares its imputed distribution with the observed distribution among FOIA PIs.

¹⁵ This restriction ties each PI's fixed effect to one institution so that institutional environments do not drive our results.

impute the exposure measure for the remaining 13,433 non-FOIA PIs.

We also link PIs in the panel to NIH RePORTER records using fuzzy name matching to measure their NIH funding. We match 7,530 of the 13,515 PIs in the panel, for whom we observe active NIH grants and annual NIH funding.

Table 3 compares PIs above and below the median exposure measure on pre-merger outcomes and characteristics. More-exposed PIs have a higher treated-market share, somewhat higher pre-merger research output, and are slightly earlier in their careers. The two groups are similar in NIH funding, active NIH grants, and the share at public universities.

Table 3. Balance Across More- and Less-Exposed PIs

	Less exposed		More exposed		Difference			
	Mean	SD	Mean	SD	Coef.	(SE)	<i>p</i> -value	N
<i>Panel A: Pre-period (2010–2013) PI averages</i>								
Publications (last author)	2.09	1.28	2.23	1.47	0.13	(0.02)	0.000	13,515
Publications (any position)	4.06	2.69	4.26	2.95	0.20	(0.05)	0.000	13,515
Citation-weighted output	4.02	4.42	4.50	5.19	0.49	(0.08)	0.000	13,515
Active NIH grants	1.25	1.00	1.29	1.05	0.04	(0.02)	0.080	7,530
NIH funding (\$ thousands)	490.52	660.39	484.73	526.28	−5.79	(13.78)	0.674	7,530
<i>Panel B: PI characteristics</i>								
Exposure measure	0.005	0.018	0.052	0.024	0.047	(0.000)	0.000	13,515
Treated-market share S_i	0.246	0.113	0.291	0.101	0.045	(0.002)	0.000	13,515
Years as PI in 2014	17.3	11.4	16.6	11.3	−0.8	(0.2)	0.000	13,515
Public university	0.61	0.49	0.61	0.49	−0.01	(0.01)	0.355	13,515
Matched to NIH records	0.55	0.50	0.56	0.50	0.01	(0.01)	0.143	13,515

Notes: One observation per PI in the all-journal R1 and R2 panel (13,515 PIs, 6,758 less-exposed and 6,757 more-exposed). More-exposed PIs are those whose exposure measure is above its median across PIs (0.021); less-exposed PIs are at or below the median. N is the number of PIs in each row. Panel A averages each outcome over a PI's pre-merger years, 2010–2013. Active NIH grants and NIH funding are observed only for PIs matched to NIH records (3,723 less-exposed and 3,807 more-exposed PIs). Panel B reports time-invariant PI characteristics. Years as PI is the number of years since a PI's first last-authored publication, measured in 2014. Public university and matched to NIH records are shares. The exposure measure and treated-market share S_i are the shift-share cost increase and the pre-merger treated-market spending share defined in Section 4.1.3, observed for FOIA PIs and imputed for the rest. Difference is the more-exposed minus less-exposed mean, with heteroskedasticity-robust standard errors in parentheses and the *p*-value of the two-sided test that it equals zero.

4.2 Average Effect on Research Output

In this subsection, we show that PIs facing larger increases in the cost of their pre-merger consumables bundle publish fewer papers after the merger. We compare changes in research output across PIs with different exposure to these price increases, controlling for their treated-market share so that the exposure measure varies only with how much prices rose in the markets from which PIs purchased.

4.2.1 Empirical Specification

Our empirical design treats each PI's exposure to the merger as a continuous treatment and estimates the effect on research output with an event study that compares the publication trajectories of more- and less-exposed PIs around the 2014 merger. PIs purchased different bundles of consumables before the merger and price increases varied across product markets, so PIs who relied more heavily on the hardest-hit markets face the largest cost increases.

We estimate an event study of publication counts that interacts each PI's exposure measure with relative-year indicators:

$$E[y_{it} \mid \mu_i, \lambda_t] = \exp \left(\mu_i + \lambda_t + \sum_{\substack{n=-4 \\ n \neq -1}}^5 \beta_n \mathbf{1}\{t - 2014 = n\} \times \text{Exposure}_i + \sum_{\substack{n=-4 \\ n \neq -1}}^5 \psi_n \mathbf{1}\{t - 2014 = n\} \times S_i \right) \quad (5)$$

where y_{it} is the raw publication count for PI i in year t . We estimate the model using Poisson pseudo-maximum likelihood (PPML). Research output is a nonnegative count with many zeros, and we are interested in proportional changes in output rather than changes in the number of papers. PPML retains observations with zero publications while allowing the coefficients to capture proportional changes in expected research output ([Chen and Roth, 2024](#)). Because the exposure measure is also expressed in proportional terms—as the percentage increase in the cost of a PI's pre-merger consumables bundle—the coefficients β_n can be interpreted as the elasticity of output with respect to consumables costs.

The terms μ_i and λ_t are PI and year fixed effects, and $\mathbf{1}\{t - 2014 = n\}$ indicates that year t is n years from 2014. We also interact the relative-year indicators with the PI's treated-market share, S_i , defined in Section 4.1.3. We omit $n = -1$, so each β_n measures the relationship between the exposure measure and research output relative to the year immediately before the merger.

A PI can have a higher exposure measure either because they spent more of their pre-merger budget in treated markets overall or because, among the treated markets they purchased from, they relied more heavily on those that experienced larger price increases. We want to isolate the second source of variation, which reflects differences in the merger-induced price shifts across markets. We therefore control for the PI's treated-market share, S_i , interacted with the relative-year indicators. This compares PIs with similar overall spending shares in treated markets but whose pre-merger purchases exposed them to different price increases. This control accounts for the fact that the spending shares entering the exposure measure do not sum to one, following [Borusyak et al. \(2022\)](#).

Our identification comes from the market-level price shifts generated by the merger, rather than from assuming that PIs' pre-merger spending shares are exogenous ([Goldsmith-Pinkham et al., 2020](#); [Borusyak et al., 2022](#)). The spending shares can reflect a PI's research field, experimental protocols, lab size, and other factors that also affect research output. The identifying assumption is instead that the merger-induced price shifts across product markets are unrelated to the counterfactual research trajectories of the PIs exposed to those markets. In other words, markets purchased disproportionately by PIs whose research output would otherwise have risen or fallen differently must not systematically experience larger merger-induced price changes. Under this assumption, the spending shares themselves may be correlated with PIs' research trajectories, provided the regression controls for their sum, which is why S_i enters equation (5).

The identifying assumption is plausible because the size of the merger-induced price increase in each market depends on the extent of pre-merger competition between Thermo Fisher

Scientific and Life Technologies. These differences are determined by the firms' pre-merger product portfolios and market structure rather than by the research trajectories of the PIs who purchased their products. Consistent with this explanation, the estimated price effects are larger in markets with greater pre-merger concentration in our procurement records ($r = 0.156$).¹⁶

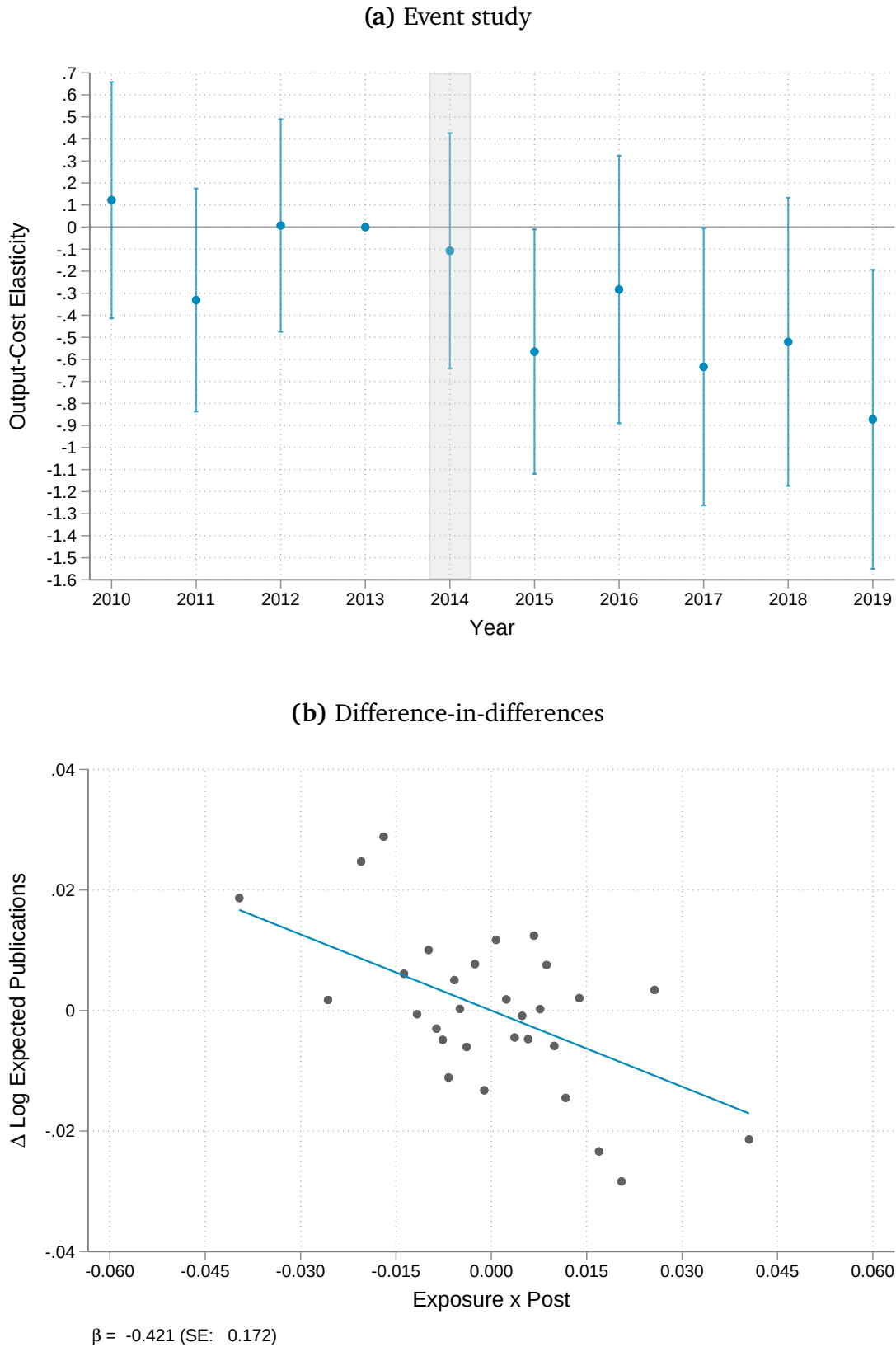
4.2.2 Results

We find that higher consumables costs reduce research output, with an elasticity of output with respect to consumables costs of approximately -0.42 . Panel (a) of Figure 6 shows the event-study estimates. The pre-merger coefficients are individually insignificant and show no systematic trend, while the post-merger coefficients are negative and become larger in magnitude over time. Thus, more- and less-exposed PIs do not appear to be on systematically different publication trajectories before the merger, but more-exposed PIs face a larger decline in research output after the merger.

Pooling the post-merger years, Table 4 reports the corresponding Poisson difference-in-differences estimates. In our preferred specification, which controls for the treated-market share, the estimated coefficient is -0.421 , compared with -0.417 without the control. The similarity of the two estimates shows that the estimated decline is not driven by differences across PIs in their overall spending share in treated markets. Panel (b) of Figure 6 reports the corresponding binned scatterplot.

¹⁶We measure pre-merger concentration using a Herfindahl–Hirschman index constructed from the procurement records. The supplier-name inconsistencies described in Section 3.2 make this measure noisy, so we view the positive correlation as supporting evidence rather than a precise estimate of the relationship between concentration and the merger's price effects.

Figure 6. Effect of Higher Consumables Prices on Publication Counts



Notes: Panel (a) plots event-study estimates from equation (5), the coefficients on the interaction of the exposure measure with relative-year indicators, with the vertical gray band indicating the merger year (2014) and coefficients normalized to zero at $t = -1$. Panel (b) is a binned scatter of annual publication counts against the $\text{Post}_i \times \text{Exposure}_i$ interaction, after partialling out PI and year fixed effects and the treated-market share control. Each point is a bin of PI-year observations. The slope of the fitted line is the difference-in-differences coefficient, $\beta = -0.421$ (SE 0.172). The sample is the panel of 13,515 PIs at 198 universities, with a pre-merger mean of 2.16 publications per PI-year.

Table 4. Effect of Higher Consumables Prices on Research Output

	(1) With share control	(2) Without share control
Exposure measure	-0.421** (0.172)	-0.417** (0.171)
Treated-market share S_i	0.007 (0.052)	
Pre-period mean	2.16	2.16
Observations	135,150	135,150

Notes: Poisson pseudo-maximum likelihood (PPML) difference-in-differences estimates of the effect of the exposure measure on annual publication counts, from a panel of 13,515 PIs observed 2010–2019 with PI and year fixed effects. The reported coefficient is on $\text{Post}_t \times \text{Exposure}_i$ and is an elasticity of output with respect to consumables costs, so the implied effect is proportional to a PI's baseline output. Column (1) adds the treated-market share control following [Borusyak et al. \(2022\)](#) and is our preferred specification, while column (2) omits it. Standard errors clustered at the PI level in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

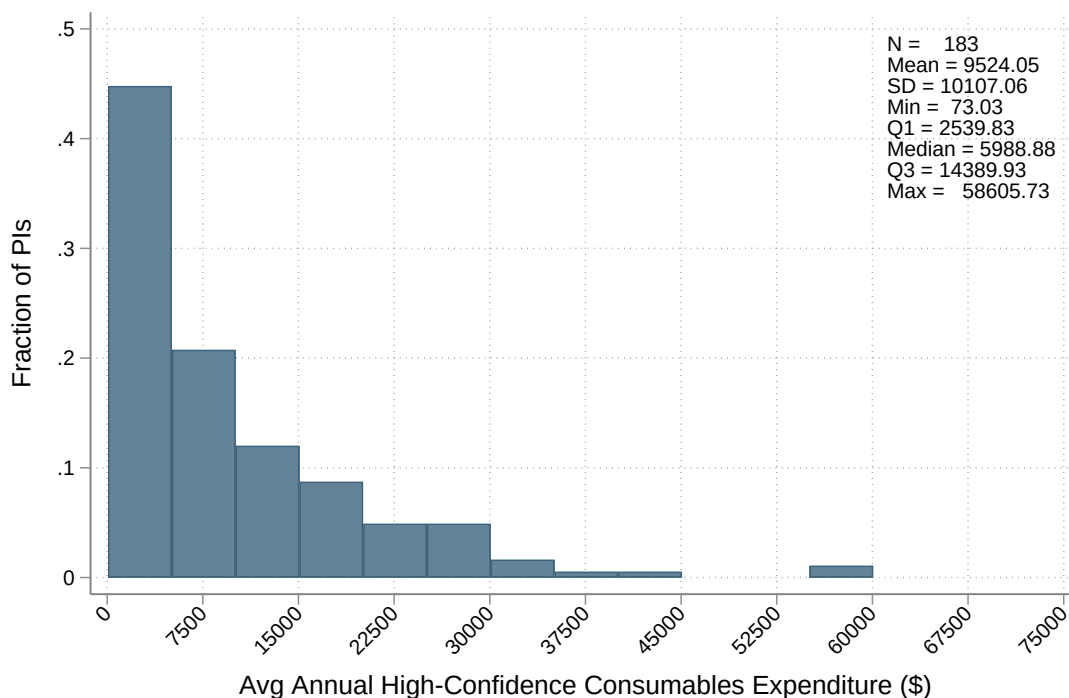
For the average PI, whose consumables costs rose by 2.8 percent, the pooled estimate implies a 1.2 percent decline in research output, or roughly 0.026 fewer papers per year.¹⁷ This elasticity is measured with respect to consumables costs alone, not the PI's full grant. Consumables account for roughly 10 to 20 percent of a grant's direct costs, and translating the estimate into a grant-level effect would require assuming that a dollar lost to higher consumables prices affects output in the same way as a dollar cut from any other research expense, an assumption our design cannot support.

To put this magnitude in dollar terms, Figure 7 shows that the average UT Dallas and UMich PI spends about \$9,500 per year in the high-confidence markets over which the exposure measure is defined, with some spending as much as \$59,000. A 2.8 percent increase in costs on this base amounts to about \$271 per year for the average PI. Combined with the estimated decline in output, this corresponds to one fewer publication for every \$10,500 increase in annual

¹⁷We interpret the coefficient as a reduced-form elasticity rather than an elasticity with respect to realized consumables spending. Recovering the latter would require estimating how the exposure measure translates into realized changes in consumables costs, which we observe only for the FOIA PIs. The current FOIA sample is too small to estimate this relationship precisely, and we are in the process of collecting additional PI-linked procurement records for this purpose. This does not pose the same limitation for the exposure imputation: the FOIA PIs serve as donors for pre-merger spending shares, which are predicted from similarities in research topics, whereas realized post-merger spending is an outcome that publication text has no reason to predict.

consumables costs.

Figure 7. Distribution of Average Annual PI Consumables Spending



Notes: The figure plots the distribution of average annual consumables spending per PI, computed from the manually classified UT Dallas and UMich FOIA records that link purchases to individual PIs, restricted to the high-confidence markets over which the exposure measure is defined. Each observation is one of the 183 PIs, with spending averaged over the years in which the PI appears in the records. Mean spending in these markets is \$9,500 per PI per year.

As a second measure of research output, we reestimate equation (5) using citation-weighted output. We weight each paper by its average annual citations and rescale the weighted counts to sum to the total number of papers, so the measure remains in units of papers and comparable to the raw count. Appendix Figure A5 reports the estimates. The estimated decline is smaller than the raw-count effect but too imprecise to distinguish from either zero or the raw-count coefficient, so we draw no conclusions about the composition of the forgone papers.

Robustness. The estimated decline is stable across alternative functional forms, fixed effects, and exposure measures. OLS estimates in levels or log publications imply declines similar to the baseline. Replacing PI fixed effects with university and research-cluster-by-year fixed effects also yields a similar decline, suggesting that differential trends across research fields do not

drive the estimate. Renormalizing each PI's spending shares across treated markets rather than controlling for the treated-market share produces a somewhat larger decline. Restricting to PIs with a positive exposure measure or weighting by text-match quality yields smaller and less precise estimates, while restricting to the closest half of imputed PIs produces an estimate near the baseline. See Appendix Tables [A5](#) and [A6](#) for these estimates.

4.3 Mechanisms

Having established that higher consumables costs reduce research output, we next ask which PIs are better able to adjust to the shock. We find that the result differs most sharply by career stage, with larger declines among early-career PIs. One possible explanation is financial resources. Late-career PIs may hold larger grants, work at better-resourced institutions, or otherwise have greater capacity to absorb higher consumables costs. We therefore examine whether the career-stage pattern is explained by differences in observable financial resources.

Among early-career PIs, the decline emerges after the merger and grows through the end of the sample, while the response among late-career PIs is smaller and imprecisely estimated, as shown in Figure [8](#). We divide PIs at the median of 14 years since their first last-authored publication as of 2014 and estimate separate exposure-by-year coefficients for early- and late-career PIs. Because last authorship identifies the PI directing the project, this measure captures experience leading an independent lab. In the pooled difference-in-differences specification, the early-career response is -0.723 (SE 0.235), compared with -0.209 (SE 0.249) among late-career PIs. The difference, -0.514 (SE 0.343), is not statistically significant at conventional levels ($p = 0.13$).

Measuring experience instead as years since a PI's first publication in any authorship position produces an even sharper split, with a gap of -0.871 (SE 0.342 , $p = 0.011$). Because this measure begins accumulating during PhD and postdoctoral training, it captures broader research experience rather than only years leading an independent lab. The two measures are highly

correlated, and both point to larger responses among less-experienced PIs.

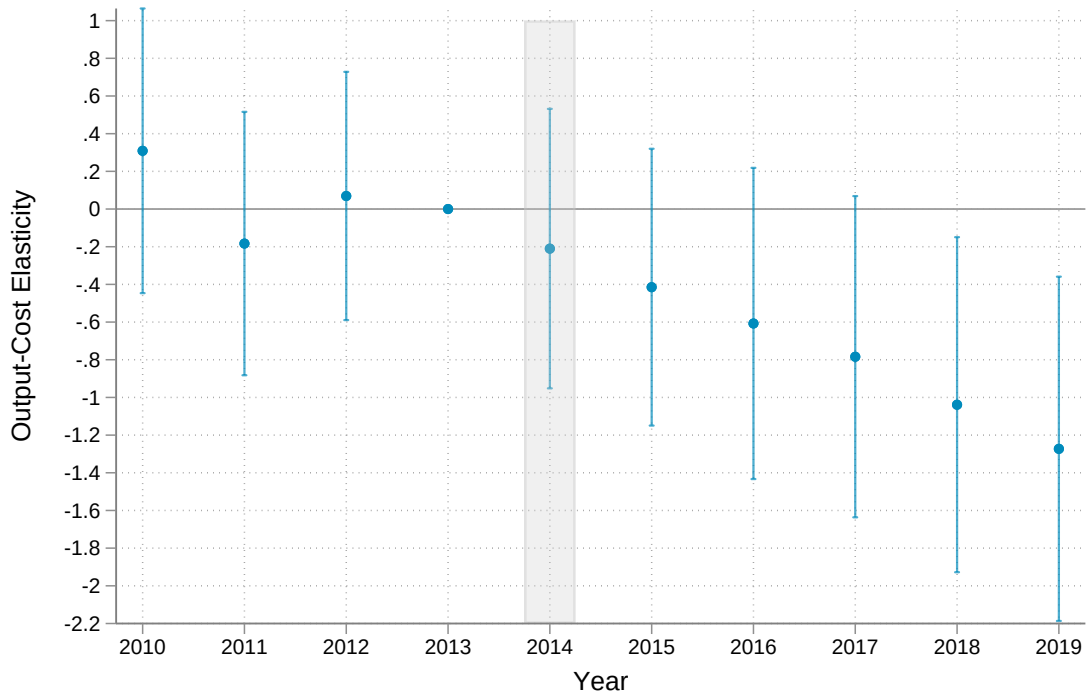
We next ask whether greater financial resources explain the career-stage pattern. Figure 9 compares the response across PIs with different levels of baseline NIH funding and across institutions that differ in research funding, endowment, and metropolitan-area size. If financial resources provide a buffer against higher input costs, we would expect smaller declines among better-resourced PIs and institutions. Instead, the estimated effect is negative in nearly every subgroup, with little evidence that better-resourced PIs or PIs at better-resourced institutions experience systematically smaller declines.

Because late-career PIs also tend to have more baseline funding, the patterns by career stage and funding may be related. We therefore compare early- and late-career PIs with similar baseline funding. Figure 10 divides PIs in the NIH-matched sample into four groups by career stage and above- or below-median pre-merger NIH funding. Within both funding groups, the point estimates are more negative for early-career PIs than for late-career PIs, although the estimates are imprecise. Thus, the career-stage pattern persists among PIs with similar baseline funding.

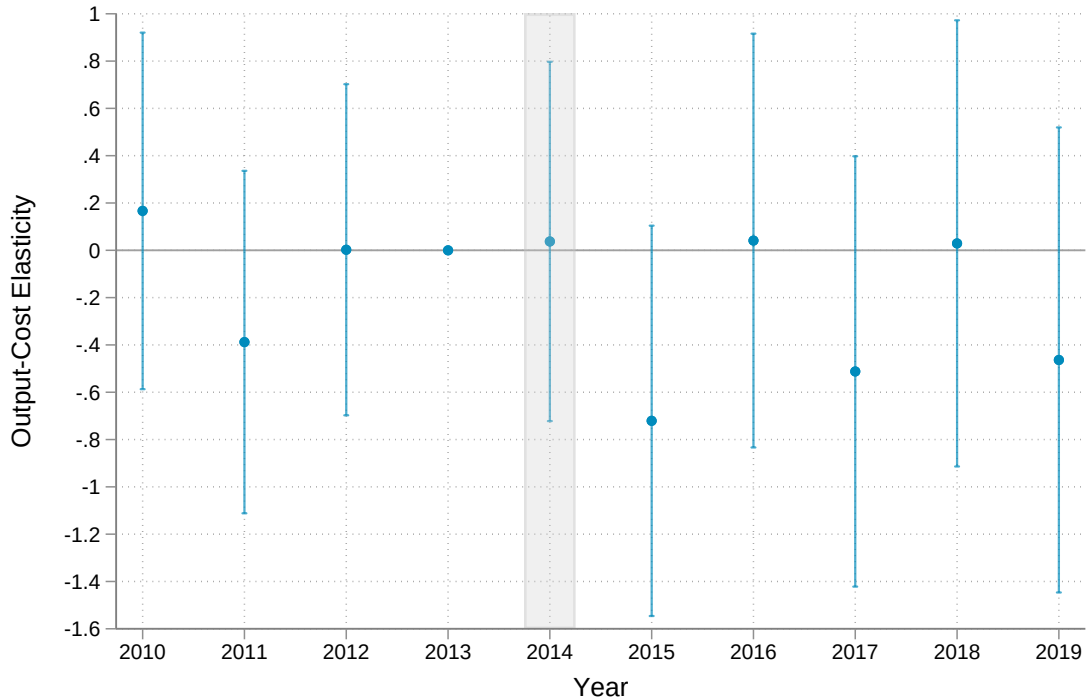
Together, these results suggest that differences in funding and institutional resources are unlikely to fully explain the larger response among early-career PIs. The pattern may instead reflect differences associated with experience, including how PIs adjust their research when input costs rise. We explore this possibility in Section 5.2 using survey evidence on how life sciences PIs manage supply costs and respond to price increases.

Figure 8. Effect of Higher Consumables Prices on Publication Counts by Career Stage

(a) Early-career PIs

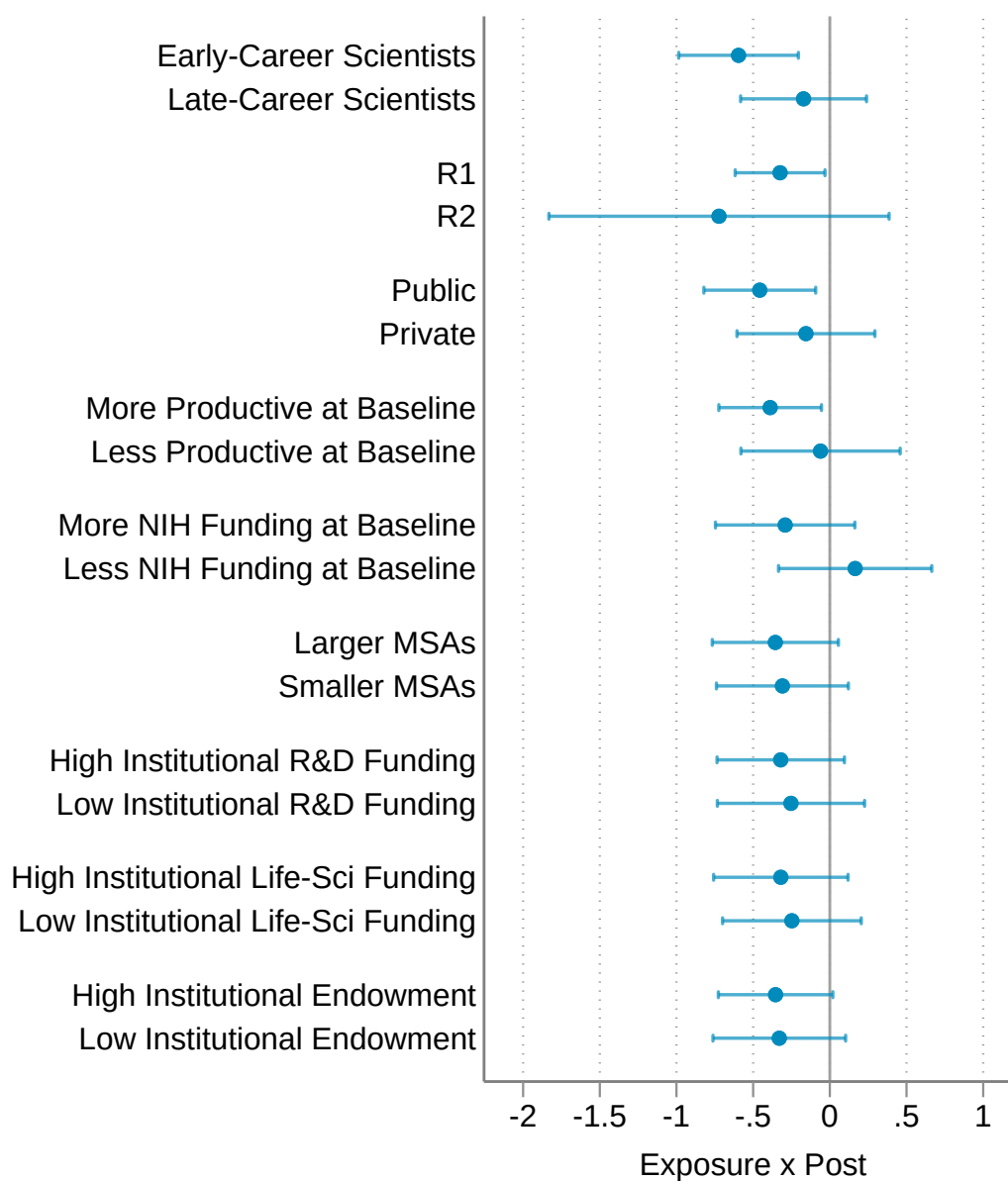


(b) Late-career PIs



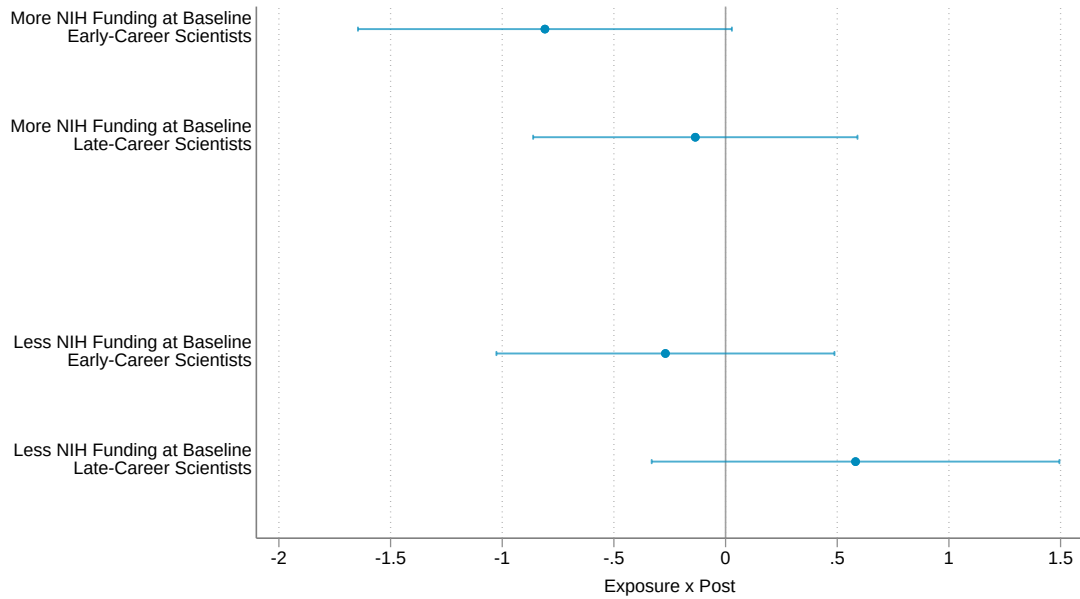
Notes: The two panels plot event-study estimates from a single regression that extends equation (5) with a separate set of exposure-by-relative-year interactions for each career stage, each panel showing one stage's coefficients, normalized to zero at $t = -1$, with the vertical gray band indicating the merger year (2014). The split is done at the median number of years (14) since a PI's first last-authored publication in 2014. Early-career PIs are those below the median (6,613 PIs at 195 universities, pre-merger mean 1.99 publications per year), and late-career PIs those above (6,902 PIs at 178 universities, pre-merger mean 2.32).

Figure 9. Effect of Higher Consumables Prices on Publication Counts by PI and Institutional Characteristics



Notes: Each row reports the Poisson difference-in-differences coefficient on $Post_t \times Exposure_i$ estimated on the panel of 13,515 PIs at 198 universities, with the two groups of each split entering one regression through separate exposure interactions. The career-stage split divides PIs at 14 years since their first last-authored publication, the panel median at 2014. The remaining rows split PIs by Carnegie classification, institutional control, and metropolitan-area size, and at the pre-merger median of baseline research output, baseline NIH funding, and institutional research and development funding, life sciences funding, and endowment. Horizontal lines are 95% confidence intervals from standard errors clustered at the PI level. Each coefficient is an elasticity of output with respect to consumables costs, a proportional effect on expected output.

Figure 10. Exposure Response by Career Stage and Baseline NIH Funding



Notes: Each point is the Poisson difference-in-differences coefficient on $\text{Post}_t \times \text{Exposure}_i$ for one of the four groups defined by career stage and below- or above-median pre-merger NIH funding, estimated in a single regression on the NIH-matched PIs with a separate set of exposure and treated-market share interactions for each group, with PI and year fixed effects. Vertical lines are 95% confidence intervals from standard errors clustered at the PI level. The above-median funding groups comprise 1,444 early-career PIs at 120 universities and 2,321 late-career PIs at 114 universities, and the below-median groups 2,123 early-career PIs at 173 universities and 1,642 late-career PIs at 151 universities, together partitioning the 7,530 NIH-matched PIs.

5 Discussion

Our results show that higher input prices reduce how much research is produced. PIs spend more on affected consumables while research output declines, with the largest declines among early-career PIs. In this section, we consider the aggregate consequences of these higher costs and why some PIs may be better able to adjust than others.

5.1 The Incidence of Higher Scientific Input Prices

Our spending and output results together show how higher input prices affect scientific production. PIs spend more on the same consumables while producing less research, so the cost of higher prices extends beyond the additional dollars paid to suppliers. The decline in output is modest for any individual PI, but because the same upstream suppliers serve PIs across many

institutions, these losses can be much larger in the aggregate.

Across the 13,515 PIs in our panel, our estimates imply approximately 349 fewer publications and \$3.7 million in additional consumables spending each year. For the average PI, this amounts to roughly \$271 in additional annual spending and 0.026 fewer publications. These estimates likely understate the effects of higher input prices for two reasons. First, our exposure measure includes price increases only in our high-confidence treated markets and assigns a zero price effect to all other treated markets. Second, measurement error in imputed exposure attenuates the estimated response toward zero. We therefore view these calculations as illustrating the scale of the effects across our panel rather than as a comprehensive estimate of the merger's welfare effects.

One fewer publication corresponds to approximately \$10,500 in higher consumables costs. By comparison, among PIs in our panel with NIH funding before the merger, annual NIH funding per publication is roughly \$100,000 for early-career PIs and \$130,000 for late-career PIs. However, NIH grants also fund personnel, equipment, and other research expenses, while consumables typically account for only about 10 to 20 percent of research budgets. Scaling the NIH benchmark by this share implies roughly \$10,000 to \$27,000 per publication in consumables spending, similar to our estimate.

These consequences may extend further downstream in the scientific innovation process. Research in the life sciences contributes to the development of new drugs and therapeutics, so reductions in research today can reduce the knowledge available for future invention. [Azoulay et al. \(2019\)](#), for example, estimate that an additional \$10 million in NIH funding generates 2.3 additional private-sector patents. While we do not estimate these downstream effects, this evidence suggests that the publication losses we observe capture only part of the consequences of higher scientific input prices.

5.2 What Allows a PI to Adjust to the Shock?

In this subsection, we argue that the larger output decline among early-career PIs points to a mechanism in which career experience helps PIs adjust, rather than to differences in financial resources. Financial resources could have explained this pattern, as late-career PIs may hold larger grants, work at better-resourced institutions, or have more slack to absorb higher consumables costs. Yet splits by various measures of financial resources show little evidence that better-resourced PIs experience smaller declines.

Experience can provide PIs with more ways to adjust when input costs rise. For instance, knowing where to save money and how to substitute materials, modify protocols, or alter the scale of experiments allows a PI to respond to a price increase without compromising the research itself. PIs develop this type of knowledge beginning in training and continuing through years of running a lab, learning from experiments and failures. Early-career PIs have had less time to accumulate it. Under this interpretation, career stage captures accumulated knowledge of how to reorganize research practices when costs increase.

To examine this mechanism more directly, we survey active U.S. life sciences PIs about how they manage supply costs and respond to price increases. Appendix E provides additional details and the full survey instrument. Data collection is ongoing, with 19 responses to date from 255 invitations, eleven from early-career PIs and eight from late-career PIs. We classify respondents who have been running their labs for fewer than 14 years as early-career and the remainder as late-career, mirroring the career-stage cutoff in our main analysis. Given the small and selected sample, we interpret the survey as suggestive evidence on the adjustment mechanisms underlying our main results.

The survey provides suggestive evidence that PIs learn to manage research costs largely through experience rather than formal training. Most respondents report learning to manage purchasing through trial and error after starting their own lab. Those who learned from others most often name a PhD or postdoctoral advisor or a lab manager, while only two respondents

report receiving institutional training. Thus, to the extent that PIs enter their careers with experience managing research costs, it appears to come primarily from time spent in other labs. Together, these responses are consistent with a learning-by-doing mechanism in which early-career PIs face cost shocks with less accumulated knowledge about how to adjust.

What PIs appear to learn through experience is how to adjust the research process itself when costs rise. Late-career PIs describe targeted changes that allow them to save costs while continuing their research. One late-career PI reports having “changed [the] method of propagating virus to save costs,” and another responds to rising prices by “plan[ning] more carefully the experiments.” A third, when pipette tips grew scarce and expensive, considered “getting a tip washer,” substituting equipment for a costly consumable.

Early-career PIs, by contrast, more often describe responding to higher costs by doing less research. One reports that key reagents have doubled in price since the lab opened in 2018 and that the lab is “constantly trying to scale down experiments to reduce cost.” Another reports running four replicates instead of six to conserve supplies, and a third writes that “[e]verything is more expensive and we can only do less.” Attempts at substitution point to the same kind of knowledge. One early-career PI “tried cheaper products, but had to switch back to the original because the cheaper ones did not work as well,” and another cannot switch certain inputs “for fear of ruining the experiment.” Knowing which changes preserve an experiment and which compromise it is the kind of knowledge that plausibly takes years of running a lab to accumulate.

Taken together, these responses provide a potential explanation for the larger output decline we estimate among early-career PIs. Although the small, self-selected sample cannot establish that accumulated knowledge drives this pattern, the responses suggest that experience can provide ways to adjust that financial resources alone do not. In this respect, PIs resemble managers, whose experience can shape how they respond to shocks ([Reagans et al., 2005](#); [Lee and Van den Steen, 2010](#); [Lamorgese et al., 2024](#)). When input costs rise, PIs must decide how to adjust the way they conduct research. Similar decisions matter for firms, where differences

in management practices help explain differences in productivity ([Bloom and Van Reenen, 2007](#); [Bloom et al., 2013](#)). Our evidence suggests that PIs become better able to make these adjustments through experience, rather than simply through having more financial resources.

6 Conclusion

How much science is produced depends partly on the firms that supply it. This paper leverages the 2014 merger of Thermo Fisher Scientific and Life Technologies to study how higher input costs affect the production of science. Using purchase-level procurement records, we show that prices rose in consumables markets where the merging firms had competitive overlap, while quantities purchased in those markets declined only temporarily. As a result, PIs spent more on these consumables. PIs more exposed to these price increases subsequently published fewer papers, with the decline concentrated among early-career PIs.

These findings quantify how supplier input markets shape how much research is produced from public research spending. When input prices rise, more grant funding goes toward purchasing the same research materials, leaving less to support research. Across the PIs in our sample, the merger increased consumables costs by an average of 2.8 percent, translating to approximately \$3.7 million in additional annual spending and roughly 349 fewer publications per year.

The results highlight the purchasing environment as a relevant factor in the design and administration of public research support. The NIH already tracks changes in the overall cost of conducting biomedical research through the Biomedical Research and Development Price Index (BRDPI). The BRDPI informs NIH budget negotiations by estimating how much funding is needed to maintain the purchasing power of research expenditures ([National Institutes of Health, 2025](#)). An aggregate index, however, captures price changes across a broad bundle of research inputs and may not reflect substantial increases within particular input markets. A sharp price increase can therefore impose significant costs on scientists who depend on a

specific input without meaningfully moving the overall index. Thus, the structure of individual supplier markets is an important complement to the BRDPI. In particular, the consequences of input price increases may depend on whether scientists can compare prices across suppliers and identify substitute products, as well as whether institutions can coordinate purchasing to reduce exposure to increases in negotiated supplier prices.

Even within the same purchasing environment, PIs differ in their ability to adjust when input costs rise. Our results and survey evidence suggest that adjusting to higher input costs depends partly on knowledge accumulated through experience rather than on financial resources. Knowing how to reorganize projects, identify substitutes, and adjust research protocols when costs rise appears to be learned largely on the job.

Running a lab requires PIs to make many decisions beyond the science itself. PIs manage research teams, allocate funding across various research expenses, and decide what to buy and from which suppliers. Yet PIs are trained primarily as scientists, with little formal training for these other responsibilities. Our survey respondents report learning to manage costs largely through trial and error after starting their own labs. Over time, PIs therefore accumulate practical knowledge about how to run a lab when resources are constrained. The larger decline among early-career PIs may reflect their having had less time to develop this knowledge when input costs rise.

Several questions remain for future work. Publications capture only one way scientists may respond to higher research costs. Higher input prices may also change which projects scientists pursue, shift research toward less resource-intensive questions, or alter the direction of scientific work. The larger decline among early-career PIs raises a separate question about longer-run effects. If early-career PIs are less able to adjust, cost increases early in a PI's career could affect future funding, career progression, or retention in science. Finally, if experience helps PIs adjust to higher costs, an important question is whether this knowledge can be passed on to other scientists rather than learned only through years of running a lab.

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A Appendix Tables and Figures

Table A1. Top Universities, Suppliers, and Product Markets by Consumables Spending

	Total Spending (\$)	% of Total
Panel A. Universities		
University of Florida	19,430,311	25.7%
University of Michigan at Ann Arbor	8,810,009	11.7%
Georgia Institute of Technology	7,286,855	9.6%
University of Kentucky	4,043,671	5.4%
Wayne State University	3,856,417	5.1%
Texas Tech University	3,509,126	4.6%
Texas A&M University	3,108,487	4.1%
Florida State University	2,978,085	3.9%
East Carolina University	2,862,949	3.8%
Washington State University	2,754,592	3.6%
Panel B. Suppliers		
Thermo Fisher Scientific	34,901,935	46.2%
VWR	12,656,994	16.8%
Life Technologies	5,290,377	7.0%
MilliporeSigma	4,248,725	5.6%
Bio-Rad Laboratories	1,946,443	2.6%
Qiagen	1,903,158	2.5%
USA Scientific	1,405,545	1.9%
Atlanta Biologicals	1,061,611	1.4%
New England Biolabs	841,844	1.1%
Genesee Scientific	679,065	0.9%
Panel C. Product Markets		
Pipette Tips	6,319,587	8.4%
U.S.-origin fetal bovine serum	5,810,406	7.7%
Serological Pipettes	4,763,075	6.3%
Vials	3,452,481	4.6%
Cell Culture Plates	2,596,718	3.4%
Petri Dishes	2,348,829	3.1%
Microcentrifuge Tubes	1,982,841	2.6%
Cell Culture Flasks	1,753,628	2.3%
Cell Culture Dishes	1,340,948	1.8%
Cuvettes	1,326,283	1.8%

Notes: Total consumables spending from 2010 to 2019 in the final estimation sample of high-confidence product markets, with shares of that sample's total. Thermo Fisher Scientific and Life Technologies are the merging firms.

Table A2. Classifier Performance on Held-Out Purchases

	Precision	Recall	N
<i>Step 1: Consumables Versus Non-consumables Classifier</i>			
Consumables	0.96	0.98	31,551
Non-consumables	0.93	0.85	8,324
<i>Step 2: Product Market Classifier</i>			
All consumables categories	0.54	0.70	295,622
Large Markets (≥ 25 training examples)	0.74	0.73	291,394

Notes: Labels come from the hand-labeled UT Dallas and UMich purchases held out from training. Step 2 conditions on true consumables and averages precision and recall equally across product markets. Large Markets restricts to markets with at least 25 training examples, which cover 98.6 percent of true consumables.

Table A3. Pooled Merger Effects on Prices, Quantities, and Spending

	(1)	(2)	(3)
Panel A: Log price			
Post $\times$ Treated	0.164* (0.099)	0.162* (0.096)	0.153 (0.096)
Panel B: Log quantity			
Post $\times$ Treated	0.023 (0.063)	0.050 (0.064)	0.001 (0.063)
Panel C: Log spending			
Post $\times$ Treated	0.199* (0.111)	0.225* (0.121)	0.167 (0.112)
University FE	✓		
Market FE	✓		✓
Year FE	✓	✓	
University $\times$ Market FE		✓	
University $\times$ Year FE			✓
Observations	19,388	19,388	19,330

Notes: Each entry is the pooled difference-in-differences coefficient on $D_m^T \times \text{Post}_t$ from a separate regression of the panel's log outcome, estimated on the high-confidence treated and matched control markets, 2010–2019. Column (1) includes university, market, and year fixed effects, column (2) university-by-market and year fixed effects, and column (3) university-by-year and market fixed effects. These unshrunk pooled estimates differ from the empirical Bayes market-level price effects $\hat{\beta}_m^p$ used elsewhere in the paper. Standard errors clustered by market in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4. Placebo-Timing Tests for the Merger Difference-in-Differences

	Placebo 2012		Merger 2014	
	$\hat{\beta}$	SE	$\hat{\beta}$	SE
Average log price	-0.034	(0.032)	0.164*	(0.099)
Log quantity	-0.070	(0.064)	0.023	(0.063)
Log spending	-0.109*	(0.064)	0.199*	(0.111)

Notes: Pooled difference-in-differences estimates on the matched treated and control markets. The placebo columns use the pre-merger panel, 2010–2013, with a false post indicator for $t \geq 2012$, and the merger columns use the full 2010–2019 panel with the post indicator for $t \geq 2014$. Both are weighted by 2013 spending and include year, university, and market fixed effects, with standard errors clustered by market. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5. Robustness of the Output Effect to Functional Form and Fixed Effects

	Poisson		OLS Levels		OLS Log	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Post × Exposure</i>						
With S_i control	-0.421** (0.172)	-0.415* (0.213)	-0.963*** (0.312)	-1.119*** (0.352)	-0.209** (0.098)	-0.195* (0.107)
No S_i control	-0.417** (0.171)	-0.324 (0.211)	-1.146*** (0.306)	-1.081*** (0.347)	-0.244** (0.096)	-0.187* (0.104)
Implied output decline (%)	1.2	1.2	1.3	1.5	0.9	0.8
<i>Fixed effects</i>						
PI + year	✓		✓		✓	
University + cluster × year		✓		✓		✓
Pre-period mean	2.16	2.16	2.16	2.16	2.16	2.16
Observations	135,150	135,150	135,150	135,150	135,150	135,150

Notes: Each column is a difference-in-differences regression of annual publication counts on Post × Exposure, with and without the treated-market share S_i as a control, following [Borusyak et al. \(2022\)](#). The OLS Log outcome is the log of one plus the count, and even columns also control for the levels of exposure and S_i . The implied output decline applies the with- S_i coefficient to the average exposure of 0.028, as a percent of the pre-period mean. Column (1) with the share control is the specification of Table 4, and all columns share its sample. Standard errors clustered by PI in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

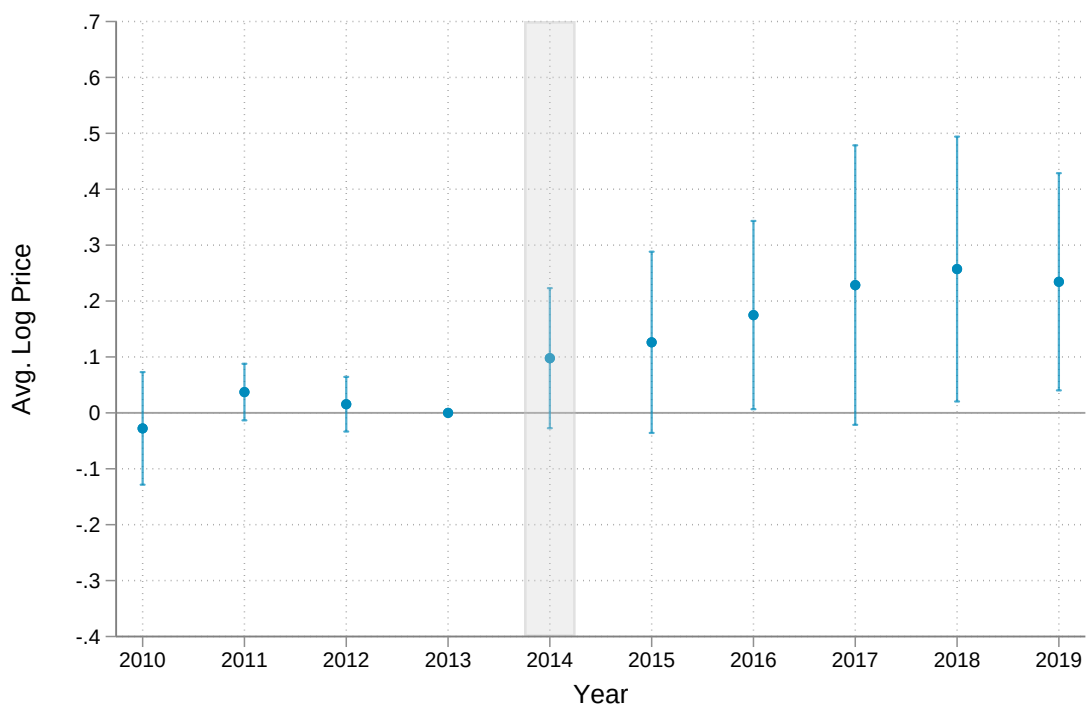
Table A6. Robustness of the Output Effect to the Exposure Measure

	(1) Baseline	(2) Renormalized shares	(3) Positive exposure	(4) Precision- weighted	(5) High- similarity
<i>Post × Exposure</i>					
With S_i control	−0.421** (0.172)	−0.139*** (0.053)	−0.252 (0.248)	−0.273 (0.177)	−0.419** (0.211)
No S_i control	−0.417** (0.171)	—	−0.289 (0.224)	−0.290 (0.177)	−0.424** (0.212)
Implied output decline (%)	1.2	1.5	0.9	0.7	1.1
Pre-period mean	2.16	2.16	2.16	2.16	2.43
Observations	135,150	135,150	120,700	135,150	67,980

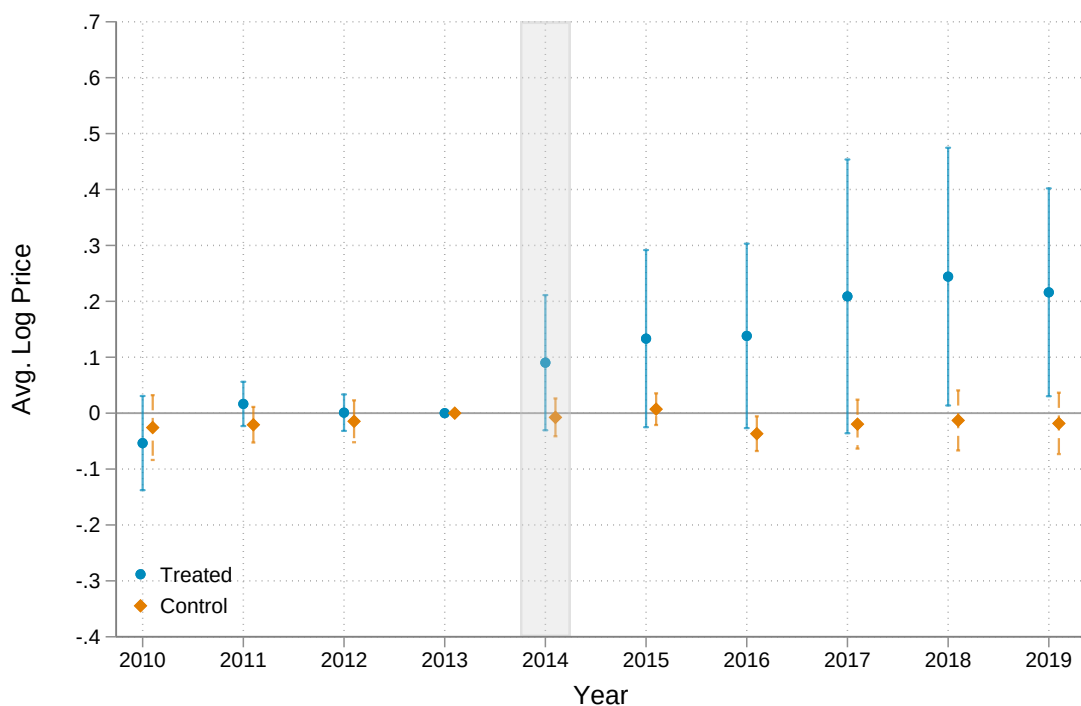
Notes: Each column is a Poisson (PPML) difference-in-differences regression of annual publication counts, as in column (1) of Table A5, varying the exposure measure. Column (2) renormalizes each PI's market shares to sum to one, the [Borusyak et al. \(2022\)](#) alternative to controlling for S_i , so the no-control row does not apply. Column (3) keeps PIs with strictly positive exposure. Column (4) weights each PI by the maximum text similarity of their publications to a FOIA PI, and column (5) keeps the top half of imputed PIs by that similarity. The implied output decline applies each column's coefficient to its average exposure, as a percent of the pre-period mean. Significance:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A1. Pooled Merger-Induced Price Effects, Naive Specification
(a) Pooled difference

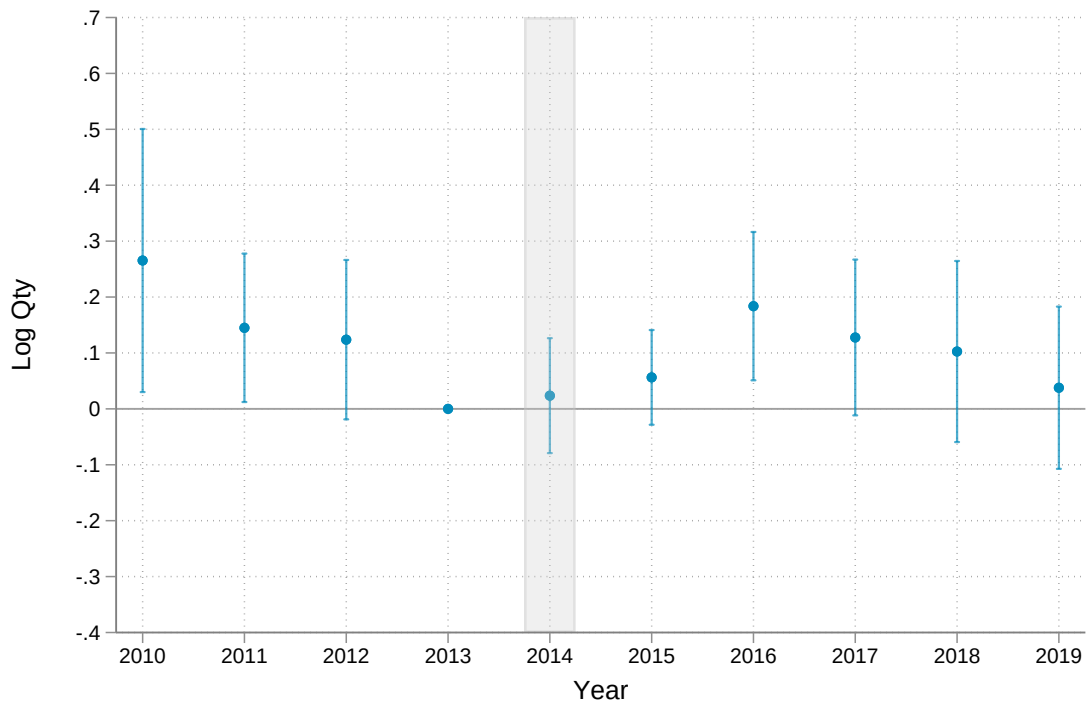


(b) Treated and control market effects

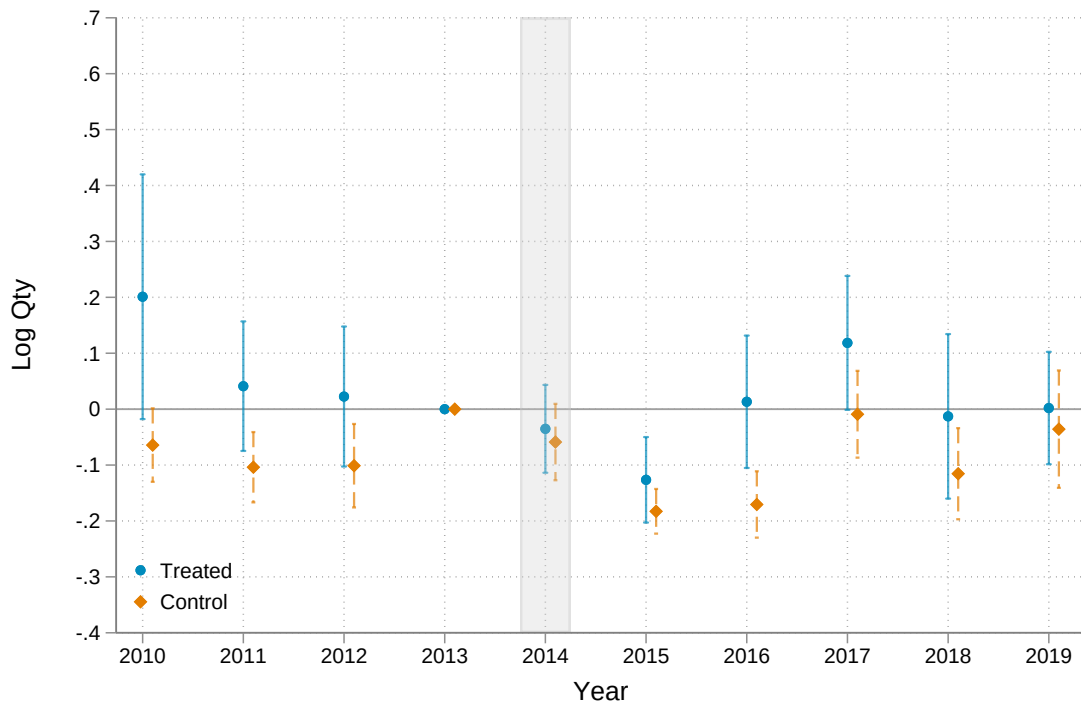


Notes: Naive event-study estimates of average log prices around the merger, using all unmatched control markets. Panel (a) plots the difference between treated and control markets and Panel (b) the two groups separately, treated (blue) and control (orange). Coefficients are normalized to zero at $t = -1$, and the gray band marks the merger year. The average price at $t = -1$ is \$143.30 in treated markets and \$84.84 in control markets.

Figure A2. Pooled Merger-Induced Quantity Effects, Naive Specification
(a) Pooled difference

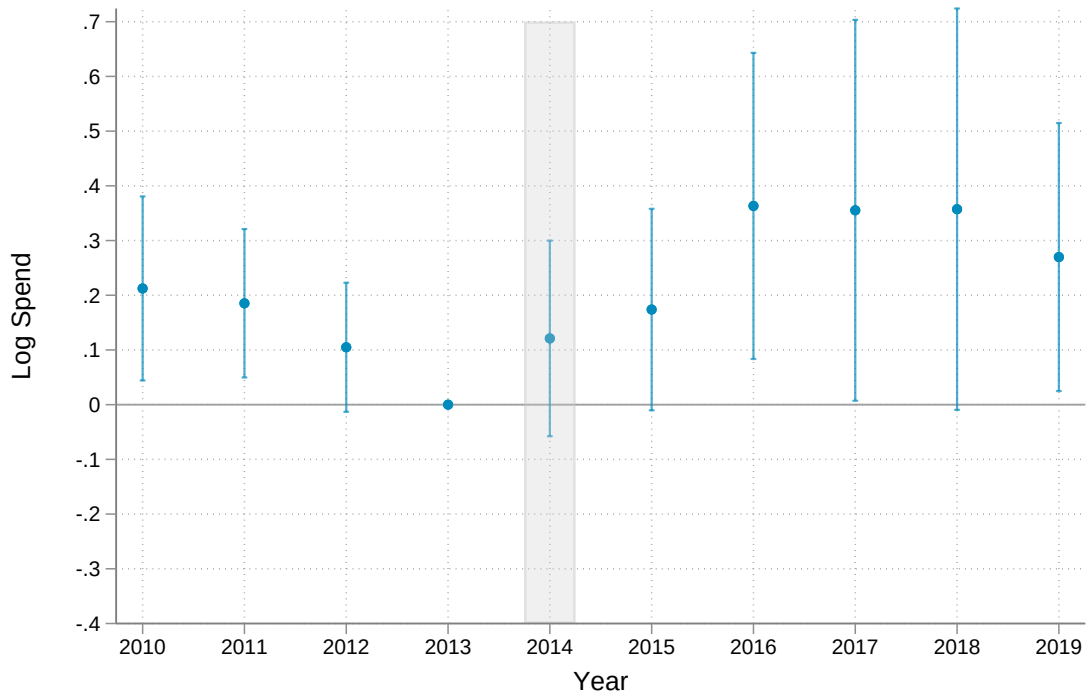


(b) Treated and control market effects

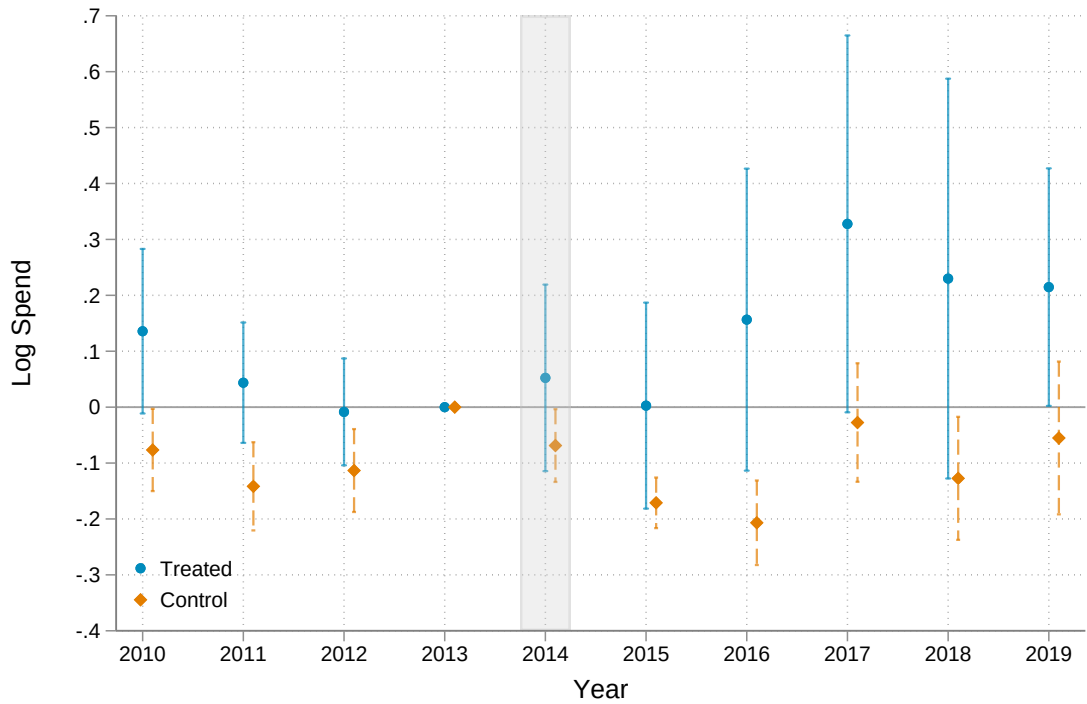


Notes: Naive event-study estimates of average log quantity around the merger, using all unmatched control markets. Panel (a) plots the difference between treated and control markets and Panel (b) the two groups separately, treated (blue) and control (orange). Coefficients are normalized to zero at $t = -1$, and the gray band marks the merger year. The average quantity at $t = -1$ is 17.8 purchases in treated markets and 30.0 in control markets.

Figure A3. Pooled Merger-Induced Spending Effects, Naive Specification
(a) Pooled difference

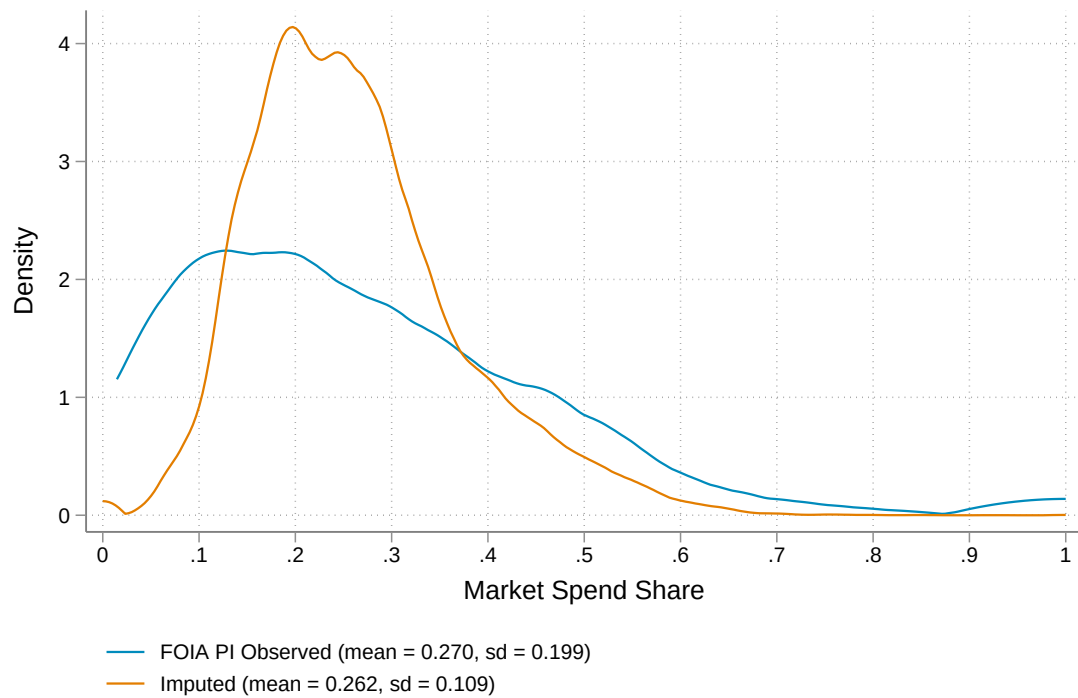


(b) Treated and control market effects



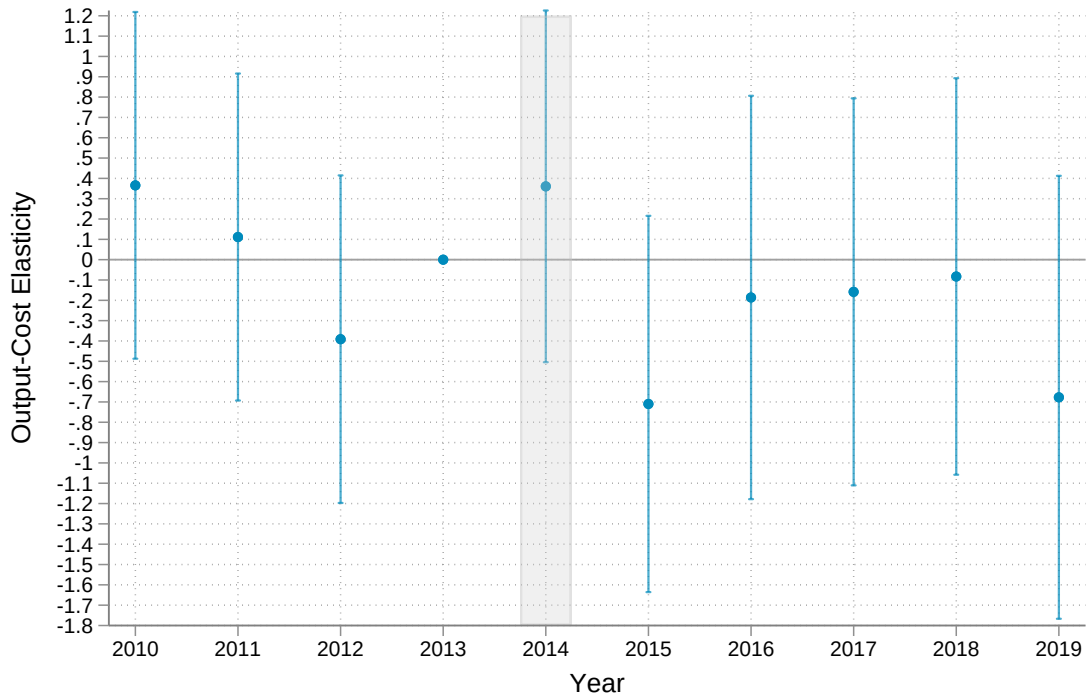
Notes: Naive event-study estimates of average log spending around the merger, using all unmatched control markets. Panel (a) plots the difference between treated and control markets and Panel (b) the two groups separately, treated (blue) and control (orange). Coefficients are normalized to zero at $t = -1$, and the gray band marks the merger year. Average spending at $t = -1$ is \$2,094 in treated markets and \$1,803 in control markets.

Figure A4. Observed and Imputed Treated-Market Share Distributions

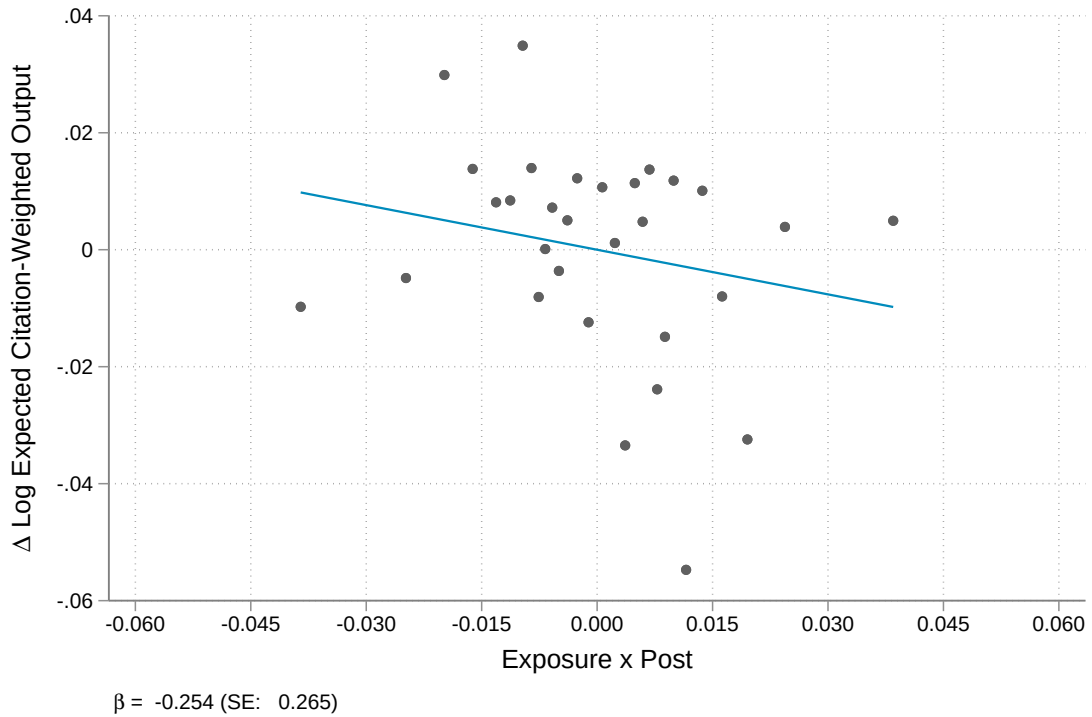


Notes: Distribution of the treated-market spending share $S_i = \sum_{m \in \mathcal{T}} s_{im}$, observed for the 208 FOIA PIs (blue) and imputed for the 13,433 non-FOIA panel PIs (orange). S_i sums a PI's pre-merger spending shares across the high-confidence treated markets and enters equation (5) as a control.

Figure A5. Effect of Higher Consumables Prices on Citation-Weighted Output
(a) Event study

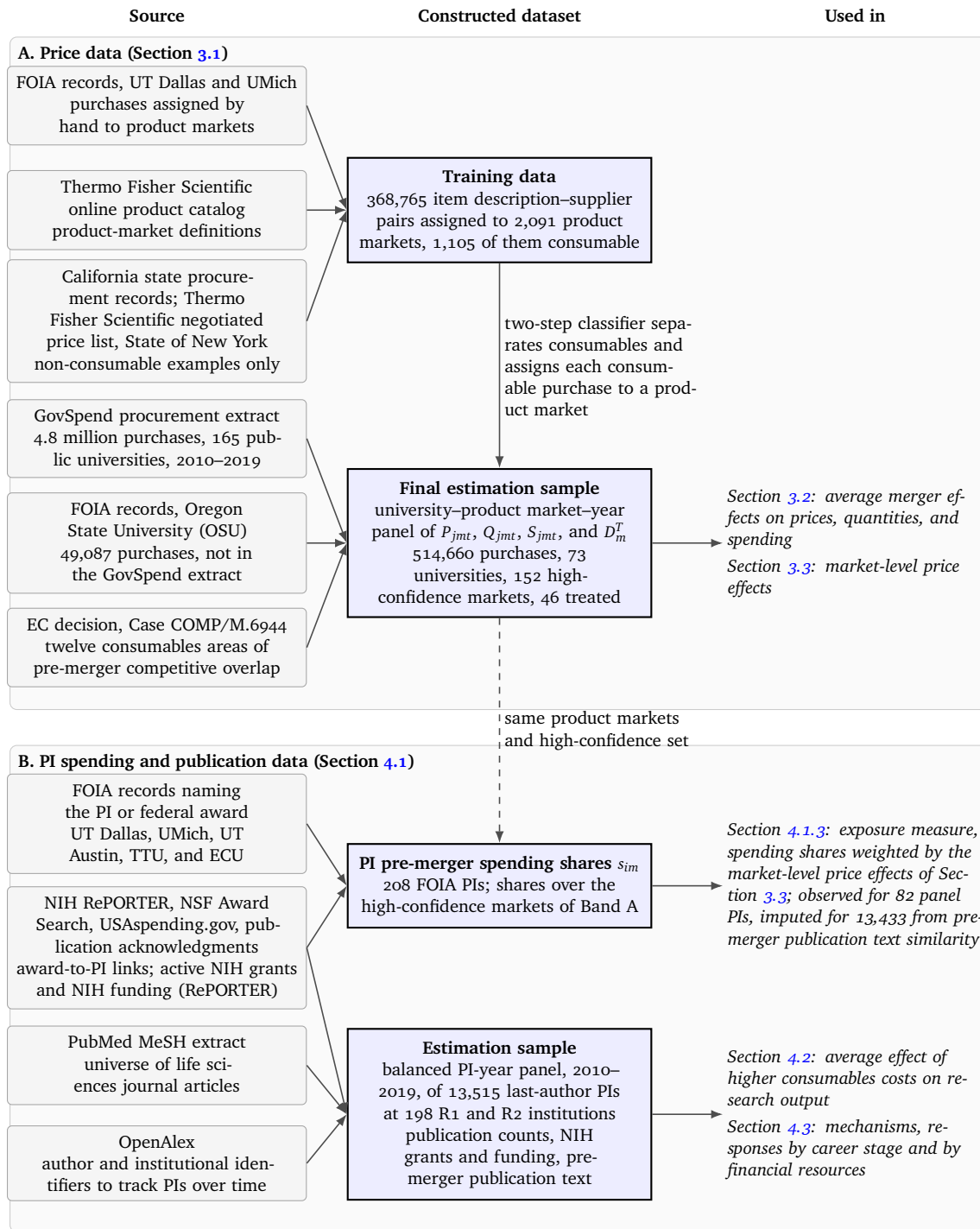


(b) Difference-in-differences



Notes: Panel (a) plots event-study estimates from equation (5) with citation-weighted output as the outcome, normalized to zero at $t = -1$. Panel (b) is a binned scatter of that outcome against $\text{Post}_t \times \text{Exposure}_i$ after partialling out PI and year fixed effects and the S_i control, with fitted slope $\beta = -0.254$ (SE 0.265). The measure weights each paper by its average annual citations and the author's effective contribution, rescaled so the weighted counts sum to the number of papers. The sample is the baseline panel of 13,515 PIs, with a pre-merger mean of 4.26 per PI-year.

Figure A6. Data Sources and Constructed Datasets



Notes: Gray boxes are data sources, blue boxes the datasets we construct from them, and the right column the section in which each is used. Band A builds the price data of Section 3.1 and Band B the PI spending and publication data of Section 4.1. The dashed arrow marks the data link between the bands. The FOIA PIs' purchases are assigned to the same product markets, by hand for UT Dallas and UMich and by the classifier for UT Austin, TTU, and ECU, and their spending shares are computed over the same high-confidence markets. Estimation steps are not shown, including the market-level price effects of Section 3.3 that enter the exposure measure.

Figure A7. Product Type Field in a Supplier Catalog Entry



Cytiva HyClone™ Defined Fetal Bovine Serum (U.S.)

Catalog No.: SH3007002HI

\$550.00

Description	Specifications	Compare Specifications	Product Suggestions	Accessories	Promotions
Additional Assays	Progesterone,Insulin,Cholesterol,HDL,LDL,Triglycerides,Gamma Globulin,Phospholipids,Alkaline Phosphatase,Lactate Dehydrogenase,SGPT,SGOT,B1,B2,B6,Niacin,Biotin,Retinol,Carotene,Tocopherol,B12,Folate,Vitamin C,1,25-dihydroxy vitamin D,pH,Total Protein,Albumin,Blood Urea Nitrogen,Creatinine,Total Bilirubin,Inorganic Phosphorus,Selenium,Sodium,Potassium,Calcium,Chloride,Magnesium,Glucose,Copper,Osmolality,Iron,TIBC,Percent Saturation,IgG				
Certifications/Compliance	Manufactured in cGMP (21 CFR 820) compliant and ISO9001 certified facilities				
Content And Storage	Sera should be stored at -10°C or lower. Once thawed, sera should be stored at 2 to 8°C.				
Country of Origin	United States				
Product Type	Fetal Bovine Serum				

Notes: Specifications page for a Cytiva HyClone defined fetal bovine serum product in Thermo Fisher Scientific's online catalog. The highlighted Product Type field, Fetal Bovine Serum, gives the initial product-market assignment. The catalog records geographic origin only in a separate Country of Origin field, so U.S.-origin and New Zealand-origin serum share one product type, whereas the European Commission's decision treats them as distinct markets. We therefore divide this product type by origin.

B Keyword Filters for the GovSpend Extract

This appendix lists the 1,505 keywords that define our GovSpend extract. The extract keeps every purchase at a public university whose item description contains one of these keywords as a substring, so *tube* also returns *centrifuge tubes* and *microtubes*. Item descriptions are free text entered by purchasing staff and vendors, so the list includes abbreviations, truncated stems, and common misspellings, such as *fltr*, *sod bicarb*, and *inocculating loop*. The keywords are meant to capture all life sciences purchases rather than consumables alone, and the classifier of Section 3.1.2 separates consumables from the non-consumable purchases they also return, such as journal page charges.

Enzymes

Restriction endonucleases and nicking enzymes

AanI	AscI	Bme139oI	BsgI	BssHII	BtgZI	Eco47I
AarI	AseI	Bme18I	Bsh1236I	BssMI	BtrI	Eco47III
AatII	AsiGI	BmeRI	Bsh1285I	BssNAI	BtsCI	Eco52I
AbsI	AsiSI	BmgBI	BshFI	BssNI	BtsI-v2	Eco53kI
Acc16I	Asp718I	Bmil	BshNI	BssSI-v2	BtsIMutI	Eco57I
Acc36I	AspA2I	BmrFI	BshTI	BssT1I	BtuMI	Eco72I
Acc65I	AspEI	BmrI	BshVI	Bst1107I	Bvel	Eco81I
AccB1I	AspLEI	BmtI	BsiEI	Bst2BI	Cac8I	Eco88I
AccB7I	AspS9I	BmtI-HF	BsiHKAI	Bst2UI	Cail	Eco91I
AccBSI	AsuC2I	Bmul	BsiHKCI	Bst4CI	CciI	EcoICRI
AccI	AsuHPI	BmyI	BsiSI	Bst6I	CciNI	EcoNI
AccII	AsuII	BoxI	BsiWI	Bst98I	CellI	EcoO109I
AccIII	AsuNHI	Bpil	BsiWI-HF	BstACI	CfoI	EcoP15I
Acil	AvaI	BplI	BslFI	BstAFI	Cfr1oI	EcoRI
AcII	AvaII	BpmI	BslI	BstAPI	Cfr13I	EcoRI-HF
AcIWI	AviII	Bpu10I	BsmAI	BstAUI	Cfr42I	EcoRII
AcoI	AvrII	Bpu1102I	BsmBI-v2	BstBAI	Cfr9I	EcoRV
AcsI	AxyI	Bpu14I	BsmFI	BstBI	Clal	EcoRV-HF
AcuI	BaeGI	BpuEI	BsmI	BstC8I	CpoI	EcoT38I
AcvI	BaeI	BpuMI	BsnI	BstDEI	CseI	EgeI
AcyI	Ball	Bsa29I	Bso31I	BstDSI	Csp6I	EheI
AdeI	BamHI	BsaAI	BsoBI	BstEII	CspAI	ErhI
AfeI	BamHI-HF	BsaBI	Bsp119I	BstEII-HF	CspCI	Esp3I
AfiI	BanI	BsaHI	Bsp120I	BstENI	CviJI	FaeI
AflII	BanII	Bsal	Bsp1286I	BstF5I	CviKI-1	Fail
AflIII	BarI	Bsal-HFv2	Bsp13I	BstFNI	CviQI	Fall
AgeI	BauI	BsaJI	Bsp1407I	BstH2I	DdeI	FaqI
AgeI-HF	BbrPI	BsaMI	Bsp143I	BstHHI	DinI	FatI
AgsI	BbsI	BsaWI	Bsp172oI	BstKTI	DpnI	FauI
AhdI	BbsI-HF	BsaXI	Bsp19I	BstMAI	DpnII	FauNDI
AhlI	Bbv12I	Bsc4I	Bsp68I	BstMBI	DraI	FbII
Ajil	BbvCI	Bse118I	BspACI	BstMCI	DraIII	Fnu4HI
AjnI	BbvI	Bse1I	BspCNI	BstMWI	DraIII-HF	FokI
AjuI	BccI	Bse21I	BspDI	BstNI	DrdI	FriOI
AleI	BceAI	Bse3DI	BspEI	BstNSI	Dril	FseI
AleI-v2	BcgI	Bse8I	BspFNI	BstPAI	DsaI	Fsp4HI
AloI	BciVI	BseAI	BspHI	BstSCI	DseDI	FspAI
AluBI	BclI	BseBI	BspLI	BstSFI	EaeI	FspBI
AluI	BclI-HF	BseCI	BspMI	BstSLI	EagI	FspI
Alw21I	BcnI	BseDI	BspOI	BstSNI	EagI-HF	GsaI
Alw26I	BcoDI	BseGI	BspPI	BstUI	Eam1104I	GsuI
Alw44I	BcuI	BseJI	BspQI	BstV1I	Eam1105I	HaeII
AlwI	BfaI	BseLI	BspQI-HF	BstV2I	EarI	HaeIII
AlwNI	BfmI	BseMI	BspTI	BstX2I	EciI	HgaI
Ama87I	BfrI	BseMII	BsrBI	BstXI	Ecl136II	HhaI
Aor13HI	BfuAI	BseNI	BsrDI	BstYI	EclXI	Hin1I
Apal	Bful	BsePI	BsrFI-v2	BstZ17I-HF	Eco105I	Hin1II
ApalI	BglI	BseRI	BsrGI	Bsu15I	Eco130I	Hin6I
ApeKI	BglII	BseSI	BsrGI-HF	Bsu36I	Eco147I	HincII
ApoI	BlnI	BseX3I	BsrI	BsuI	Eco24I	HindII
ApoI-HF	BlpI	BseXI	BssAI	BsuRI	Eco31I	HindIII
ArsI	BmCAI	BseYI	BssECI	BtgI	Eco32I	HindIII-HF

HinfI	MboII	Nb.BtsI	PceI	PspPI	SfaNI	TaqI
HinP1I	MfeI	Nb.Mva1269I	PciI	PspPPI	SfcI	TaqI-v2
HpaI	MfeI-HF	NciI	PciSI	PspXI	SfiI	TaqII
HpaII	MhlI	NcoI	PcsI	PsrI	SfoI	TasI
HphI	MlsI	NcoI-HF	PctI	PstI	Sfr274I	TatI
Hpy166II	MluCI	NdeI	PdiI	PstI-HF	Sfr303I	TauI
Hpy188I	MluI	NdeII	Pdml	PstNI	Sful	TfiI
Hpy188III	MluI-HF	NgoMIV	PecI	PsuI	SgeI	Tru1I
Hpy8I	Mly113I	NheI	PfeI	PsyI	SgfI	Tru9I
Hpy99I	MlyI	NheI-HF	Pfl23II	PvuI	SgrAI	TscAI
HpyAV	MmeI	NlaIII	PflFI	PvuI-HF	SgrBI	TseFI
HpyCH4III	MnlI	NlaIV	PflMI	PvuII	SgrDI	TseI
HpyCH4IV	Mox20I	NmeAIII	PfoI	PvuII-HF	SgsI	Tsp45I
HpyCH4V	Mph1103I	NmuCI	PI-PspI	RgaI	SlaI	TspDTI
HpyF10VI	MreI	NotI	PI-SceI	RigI	SmaI	TspGWI
HpyF3I	MroI	NotI-HF	PinAI	RsaI	SmiI	TspMI
HpySE526I	MroNI	NruI	PkrI	RsaNI	SmiMI	TspRI
HspAI	MroXI	NruI-HF	Ple19I	RseI	SmlI	Tth111I
I-CeuI	MscI	NsbI	PleI	Rsr2I	SmoI	Van91I
I-SceI	MseI	NsiI	PluTI	RsrII	SnaBI	Vha464I
KasI	MslI	NsiI-HF	PmeI	SacI	SpeI	VneI
Kpn2I	Msp20I	NspI	PmlI	SacI-HF	SpeI-HF	VspI
KpnI	MspA1I	NspV	PpsI	SacII	SphI	WarmStart
KpnI-HF	MspCI	Nt.AlwI	Ppu21I	SalI	SphI-HF	Nt.BstNBI
KroNI	MspI	Nt.BbvCI	PpuMI	SalI-HF	SrfI	XagI
Ksp22I	MspJI	Nt.Bpu10I	PscI	SapI	Sse9I	XapI
KspAI	MspR9I	Nt.BsmAI	Pse31I	SatI	SseBI	XbaI
KspI	MssI	Nt.BspQI	PshAI	Sau3AI	SsiI	XceI
Kzo9I	MteI	Nt.Bst9I	Psil	Sau96I	SspDI	XcmI
LguI	MunI	Nt.BstNBI	Psil-v2	SbfI	SspI	XhoI
LmnI	Mva1269I	Nt.CviPII	Psp124BI	SbfI-HF	SspI-HF	XhoII
Lsp1109I	MvaI	OliI	Psp1406I	ScaI	SspMI	XmaCI
LweI	MwoI	PacI	Psp5II	ScaI-HF	SstI	XmaI
MabI	NaeI	PaeI	Psp6I	SchI	StuI	XmaJI
MaeIII	NarI	PaeR7I	PspCI	ScrFI	StyD4I	XmiI
MalI	Nb.BbvCI	PagI	PspEI	SdaI	StyI	XmnI
MamI	Nb.Bpu10I	PalAI	PspGI	SduI	StyI-HF	ZraI
MauBI	Nb.BsmI	PaqCI	PspLI	SetI	SwaI	ZrmI
MbiI	Nb.BsrDI	PasI	PspN4I	SexAI	TaaI	Zsp2I
MboI	Nb.BssSI	PaulI	PspOMI	SfaAI	TaiI	

Polymerases, ligases, nucleases, proteases, and other enzymes

collagenase	elastase	kinase	Mutanolysin	rnase	transferase
dehydrogenase	enzyme	liberase	peroxidase	t4	trypsin
deoxyribonuclease	exonuclease	ligase	phosphatase	t4 dna	WarmStart
dipase	hi-fi	luciferase	polymerase	t5	
dispase	high-fidelity	luciferin	protease	t7	
dnase	hot-start	lysozyme	proteinase	taq	

Reverse transcription

improm-ii	rev transcription	rev transcriptase	reverse transcriptase	reverse transcription	superscript
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Antibodies, Proteins, and Standards

Antibodies, conjugates, and antibody targets

affinipure	anti-il1a	Anti-Rat	hif-1	il-1	pAb
anbtI-bcl2	Anti-Mouse	Anti-Sheep	hrp	lc3b	polyclonal
anti-elf3	anti-mu	anti-sox9	iga	mAb	sglt2
Anti-Goat	anti-rab	antibody	igg	mcl-1	tnem173
Anti-Guinea Pig	anti-rabb	b-amyloid	igm	monoclonal	
Anti-Human	Anti-Rabbit	bamyloid	igy	nfkb	

Proteins, peptides, and recombinant products

albumin	casein	growth factor	recombinant human	streptavidin
avidin	collagen	hemin	recombinant mouse	wheat germ
beta-actin	cytochrome	peptide	reombinant	agglutinin
bsa	gelatin	protein	rhil	

Ladders and standards

bp ladder	icp standard	ladder gel	pre-stain	protein ladder	unstained protein
dna ladder	kb plus	nativemark	precision plus	protein standard	
generuler	ladder dna	Pageruler	prot std	spectra	

Nucleic Acids and Cloning Reagents

Nucleic acids, oligonucleotides, and nucleotides

adenosine 5 triphosphate	cytosine deoxynuc	dna dntp	hexamer mirna	primer ribonuclease	rna ladder
atp	deoxynucleotide	dsdna	nucleic acids	ribonucleic acid	RNAi
cdna	deoxyribonucleic acid	glycogen	oligo	rna	shrna
					siRNA

Vectors, competent cells, and cloning reagents

chem comp	comp e.coli	iptg	plasmid	x-gal	
comp cell	competent cell	one shot	t-vector		
comp e coli	competent e coli	pgem	top10		

Nucleic acid purification kits and RNA reagents

ampure	genejet	mini-prep	rna-protect	rrasin	zyzo
dneasy	maxi-prep	mprep	rnaclean	suprase	
ez-10 column	megascript	nucleospin	rnaseout	trizol	
gel extraction	mid-prep	ribolock	rnasezap	trizol reagent	
gel-pcr	minelute	ribominus	rneasy	zeoprep	

Stains, Dyes, Probes, and Indicators

acridine orange	dab	fam-mgb	hematoxylin	pH	sudan black
alcian blue	dapi	fast green	hoechst	ph strip	sybr
alexa fluor	Diamidino-2-	FITC	lysotracker	ponceau	sybr-green
aniline blue	phenylindole	fluor	mitosox	protein stain	xypro
biebrich scarlet	dye	fluorescein	neurotrace	ram stain	texas red
brilliant violet	em stain	fluoro gel	nile red	rhodamine	toluidine blue
coomassie	eosin	fluoromount	orange ii	rose bengal	trypan blue
cresyl violet	eosin y	gel stain	perox test	silversnap	
crystal violet	ethidium bromide	golgistain	peroxide test	stain	

Buffers, Chemicals, and Antibiotics

Buffers, salt solutions, and water

bis-tris	edta	extraction solution	sds	tris acetate edta	trishydrox-
buffer	egs	hepes	tae	tris base	metaminometh
chaps	eliza buffer	mes	tbe	tris borate edta	water molecular
deionized water	ethlenediamintetace	nuclease-free water	tbe pp	tris glycine	water molecular bio
diluent	ethlenediamin-	pbs	tbs	tris glycine sds	
dpbs	tetacetic	RIPA	tris	tris-hcl	

Solvents and organic reagents

2-propanol	chloroform	ethano	glycerol	paraformaldehyde	tetrahydrofuran
acetone	dimetformamide	ethanol	hexanes	peg	xylene
acetonitrile	dimethyl-sulfoxide	ethyl acetate	isopropanol	phenol	
alcohol proof	dimethylformamide	ethyl alcohol	methanol	poly ethylene glycol	
alcohol reagent	dmf	formaldehyde	methyl-sulfoxide	methyl ether	
butyllithium	dmsO	formldhyd	methylene chlor	polyethylene glycol	

Acids and bases

acetic ac	boric acid	hydrochlor ac	phosphoric acid	sodium hydroxide	trifluoroacetic ac
ammonium	citric acid	hydrochloric acid	potassium hydroxide	sulfuric ac	trifluoroacetic
hydroxide	formic acid	nitric acid	sodium hydroxd	sulfurif ac	

Salts, sugars, and amino acids

adenine	d-glucose	mag sulf	sod bicarb	sodium carbonate	sodium pyruvate
aluminum chloride	d-mannitol	magnesium chloride	sod chloride	sodium chloride	sucrose
ammonium acetate	glucose	magnesium chloride	sod hydride	sodium citrate	zinc chloride
ammonium chloride	glycine	pot carbonate	sod phos	sodium dodecyl	
calcium acetate	l-cysteine	pot phosphate	sod phosphate	sulfate	
calcium carbonate	l-glutamine	potassium carbonate	sodium acetate	sodium hydride	
calcium chloride	lithium chloride	potassium chloride	sodium azide	sodium nitrite	
d-fructose	lysine	potassium phosphate	sodium bicarbonate	sodium phosphate	

Antibiotics, antimycotics, and disinfectants

amphotericin	carbenicillin	g418	metronidazole	streptomycin	trimethoprim
ampicillin	cefoxitin	geneticin	neomycin	plasmocin	vancomycin
antimycin	ceftriaxone	gentamicin	nigericin	puromycin	zeocin
antimycotic	chloramphenicol	gentamycin	normocin	spectinomycin	
bacitracin	ciprofloxacin	germicide	nystatin	streptomycin	
blastidin	clindamycin	hygromycin	pen-strep	streptozotocin	
bleach	daptomycin	kanamycin	penicillin	tetracycline	
bleomycin	doxycycline	levofloxacin	penicillin-	tobramycin	

Other reagents and small molecules

actinomycin	dmog	hydrogen perox	mlclr sieves	protease inhib	triton
amide	doxorubicin	igepal	mole sieves	silica	tween
anisomycin	drierite	imidazole	molecular sieve	silica gel	urea
biotin	dtc	inositol	nhs ester	tcep	
carbon dioxide	edc	lectin	oligomycin	tpen	
deuterium oxide	folic acid	lipopolysaccharide	percoll	tris 2 carboxyethyl	
dextran	glutathione	maldi matrix	phosphatase inhib	phosphine	
digitonin	hbtu	mitomycin	pma	hydrochloro	

Cell Culture Media and Reagents

Basal media and balanced salt solutions

brff-hpc1	eagle s	hank balanced	megm	modified eagle	sf-900
dmehigh	earles	hbss	mem neaa	medium	stemprom
dmem	emem	hibernate a	mem non-essential	neurobasal	
dmemw	f-12 nutrient	insect medium	minimum essential	opti-mem	
dmm	gibco	keratinocyte sfm	medium	rpmi	
dulbecco	ham s	leibovitz s		serum-free	

Sera and supplements

amino acid	calf serum	fbs	fetal bovine serum	horse serum	matrigel
b-27	donkey serum	fbs hi	glutamax	hyclone	usda tested serum
bovine growth serum	fb essence	fbs usda	goat serum	hyq	vitamin supplement

Microbiological media

agar	casamino	horse blood	mueller hinton	soc medium	yeast
brain heart infusion	chromagar	lb broth	peptone	tryptone	
broth	enterococcus	lb media	sheep blood	tryptose	

Cell culture, dissociation, and transfection

accutase	cell culture	cell dissociation	tc	transfection	
cell	cell detachment	lipofectamine	tis cult	transfection reagent	
cell cult	cell dissoc	nucleofector	tissue culture	turbofect	

Assays, Kits, and Techniques

Techniques and workflows

apoptosis	dna assembly	mutagenesis	qpcr	synthesis	
cell lysis	ligation	pcr	rt qpcr	terminal labling	
cloning	lysis	purification	sequencing	terminl labling	

Assays and detection kits

alphascreen	suppressor	celltiter	duoset	pierce protein	bromide
assay	bca	colorimetric kit	elisa	premo	tsa plus
assay ki	bradford	colorimetric kt	eliza kit	protein assay	xenolight
assay kit	bright clo	deadend system	live cell	sigmastar	
assay kt	bright-glo	det kit	live/dead	smartflare	
assy	cell proliferation	detection kit	MTT	thiazolyl blue	
background	cell viability	duolink	pierce	tetrazolium	

Electrophoresis and blotting

acrylamide	electrophoresis	mini-gel	north2south	restore plus	western blot
agarose	gel loading	mini-trans	nupage	supersignal	western blotting
ammonium	gel system	n tetramethylethyle	polyacrylamide	temed	
persulfate	immobilon	nativepage	protean	tetramethylethyle	
blotting membrane	instapage	nitrcl	protogel	trans-blot	
clarity western	laemmli	nitrocellulose	pvd membrane	transfer-buf	

Purification, chromatography, and concentration

amicon ultra cncfrugl	concentrator	float-a-lyzer	magnetic bead	sephadex	uplc column
amicon	db myone	guard column	myone	sepharose	vivaspin
bio-gel	dynabead	hiload	nap 25	slide-a-lyzer	vivaspinturbo
chromatography	dynamag	hispur	ni-nta	slide-a-lyzr	zorax
column	filter column	histrap	polysep	spin column	zorbox
concentrator	filter columnhiprep	hitrap	rediseq	superdex	

PCR master mixes and other branded reagents

iscript	mannheim	omnimix	super-mix
itaq	master-mix	powerupsybrgreen	

Labware and Consumables

Glassware, vessels, and general labware

beaker	crystallizing	funnl	reagnt reservoir	stir-bar	wgh boat
bottle	cuvette	graduated cylinder	reservoir rgnt	stirbar	
btl	cyl grad	lab scoop	scoopula	stirring bar	
btle	cylinder ml	micro spatula	spin-bar	stop-cock	
combitip	finntip	ppfunl	spoon spatula	weigh boat	
cork-ring	funnel	reagent reservoir	spoonula	weigh dish	

Pipettes and pipette tips

96 rack	eptips	lr tip	pipette tip	stripet	tip str fltr
barrier tip	filter tip	macro tip	pipettor	strpet	tip-rack
beveltdip	fisherbrand elite	maxi tips	pipt tp	strpette	trans ppt
beveled-tip	fltr tip	microtip	ppettor	tip bulk	xl tip
bevtip	gel tip	pipet	refill tip	tip filter	
channel-pipette	graduated tip	pipet tip	serological	tip fltr	
channelpipette	low reten	pipetman	sftfitl	tip rack	
cliptip	low retention	pipette	softfitl	tip ster fltr	

Tubes, racks, and cryogenic storage

capillary tube	cryo fiberbox	culture tube	microcentrifuge	safe-lock	tube strip
cardboard freezer	cryobox	emr tube	microfuge	scintillation vial	tube strp
box	cryodot	epp	microtube	tb	vial
cent tbe	cryoelite	eppendorf	micrtb	tb cnt	vial cryogenic
cent tube	cryolabel	epr tube	mltbe	test tube	vial scint
centrifuge tube	cryotube	lobnd	nmr tube	tube conic	
conicl tbe	cryovial	mcrtb rck	nunc conicl	tube grad	
cryo	cult tube	mct	safe-lck	tube rack	

Plates, dishes, flasks, and slides

96wl	cover glass	flask	microslide	plate tc	tissue dish
cap strip	coverslip	flsk	multidish	plt	top seal
cell dish	cul dish	gls cv	nunc	plt 96	well plate
cellstar	culture dish	micro-slide	pcr plate	slide chamber	well plt
chamber sld	culture insert	microscope slide	pcr well	slide superfr	well-plate
chamber slide	culture plate	microscop sld	petri	tc dish	well-plt
clearseal	culture well	microscpe slide	plate 96	tc plate	
cov glass	cvr glas	microseal	plate flat	tc-flsk	

Filters, syringes, and needles

aspirator pes	extract thimble	fltr ppr	puradisc	syringe	thimble sgl
bottle top filter	filter paper	fltrbt	stericup	syringe fil	whatman
centrifugal	filter ppr	fltrbtpr	syr	syringe filter	
centrifugal filter	flt	millipak	syr flt	syrng	
centrifugal filter unit	fltr	milx	syr fltr	symre	
cncfrgl	fltr paper	needle	syr-filt	tb syr	

Histology, protective wear, and other consumables

aluminum foil	embedding cassette	histosette	latex bulb	pap pen	tissue tek
antifade	embedding medium	inoc loop	lysing matrix	paraffin	weighing paper
cell scraper	face-mask	inoculating loop	mask	parafilm	weighing ppr
cell strainer	face-msk	inoculating loop	mountant	rubber bulb	wrapping film
embed medium	glove	kimwipe	mounting media	thermal paper	
embed resin	glv	lab-coat	mounting medium	tissue grinder	

General Terms and Units

configurationid	gene	nmol	ppt
femto	hydrochloride	pmol	umol

C Treated and Control Product Markets

This appendix lists the 152 high-confidence product markets that enter the price estimation, 46 treated and 106 control, after the 34 control markets with contemporaneous shocks described in Section 3.1.3 are removed. A market is high-confidence if it has at least 25 training examples and precision and recall of at least 0.8.

A market is treated on one of two grounds. It is *directly documented* when the European Commission's clearance decision (Case COMP/M.6944) names it, or the product family containing it, as an area of pre-merger competitive overlap between Thermo Fisher Scientific and Life Technologies. It is a *complement* when the decision does not name it but it is purchased alongside a documented market in the same experimental workflow, so that the merged firm could plausibly reprice it as part of the bundle. Table A7 maps each treated market to the decision's broader product market, with the documented markets in Panel A and the complements in Panel B. The weakest anchor is site-directed mutagenesis, which attaches to the cloning workflow rather than to a named market. Robustness estimates restrict treatment to Panel A. Control markets fall outside the merging firms' overlap and are listed by category after the table.

Table A7. High-Confidence Treated Markets Mapped to Case COMP/M.6944

Product market in the decision	Our product markets
Panel A: Directly documented markets	
Media for cell culture and process liquids	<i>Media:</i> dmem; dmem/f-12; rpmi. <i>Process liquids:</i> phosphate-buffered saline (pbs) buffer; dulbecco's phosphate-buffered saline (dpbs) buffer; hanks' balanced salt solution (hbss) buffer; tris-edta (te) buffer
Sera for cell culture	us fbs; donkey serum; goat serum; horse serum
Nucleic acid amplification reagents	<i>Enzymes:</i> taq polymerases; high-fidelity dna polymerase; hot start pcr systems; reverse transcriptase. <i>Kits:</i> first-strand cdna synthesis systems
Nucleic acid purification	<i>Molecular weight standards:</i> pre-stained protein ladders; unstained dna ladders
Cloning	<i>Enzymes:</i> dnase i; rnase inhibitors; klenow fragment; t4 polynucleotide kinase
SDS-PAGE	pre-cast bis-tris gels; pre-cast tbe gels; pre-cast tris-glycine gels; pre-cast tris-tricine gels; nucleic acid gel stains; protein gel stains
Western blotting	nitrocellulose blotting membranes; pvdf blotting membranes; chemiluminescent substrates
Panel B: Complements of documented markets	
Media for cell culture	cell culture dissociation reagents; tryptone; yeast extract
Sera for cell culture; Western blotting	bovine serum albumin
Nucleic acid purification	column-based dna plasmid miniprep, midiprep, and maxiprep kits; gel dna extraction kits; microbial dna purification kits; rna stabilization reagent
Cloning	site-directed mutagenesis systems

Notes: The left column gives the product market as defined in the European Commission's clearance decision, Case COMP/M.6944 (Thermo Fisher Scientific/Life Technologies, 26 November 2013), and the right column lists our finer product markets within it. Panel A markets are named in the decision, directly or through their product family. Panel B markets are purchased alongside a Panel A market within the same experimental workflow. The purification-kit complements also draw on the kit-level overlap in the predecessor decision, Case COMP/M.5264 (Invitrogen/Applied Biosystems, 11 November 2008). Robustness estimates restrict treatment to Panel A.

Control Markets

- **Bulk chemicals, acids, and solvents.** Acetone, chloroform, ethyl acetate, isopropanol, methanol, tetrahydrofuran (THF), trifluoroacetic acid (TFA), xylenes, phenols, glycerol, glycine, nitric acid, sulfuric acid, boric acid, hydrogen peroxide, bleach, ammonium chloride, ammonium persulfate, lithium chloride, magnesium chloride, potassium carbonate, potassium phosphate, sodium acetate, sodium azide, sodium bicarbonate, sodium chloride, sodium citrate, sodium hydroxide, sodium sulfate, and zinc chloride.
- **Buffering agents.** HEPES buffer, MES buffer, and MOPS buffers.
- **Antibiotics and selection reagents.** Ampicillin, carbenicillin, chloramphenicol, hygromycin, penicillin-streptomycin, puromycin, zeocin, and IPTG.
- **Detergents, lysis, and reducing reagents.** Digitonin, SDS (sodium dodecyl sulfate), Triton X-100, Tween detergents, DTT, TCEP, and protease inhibitor cocktails.
- **Stains, substrates, and detection kits.** Apoptosis detection kits, bioluminescence assay kits, colorimetric substrates (DAB, MTT, and TMB), histology and cell stains, phalloidin conjugates, viability stains, western blot stripping buffers, and PAP pens.
- **Separation media and other molecular-biology reagents.** Agarose, dNTPs, GST-tag affinity resins, size-exclusion resins, spin desalting columns, and dialysis cassettes.
- **Cell culture and microbiology reagents.** Carbon dioxide, laminin, polylysines, thrombin, dehydrated bacterial broth, and inoculating loops and needles.
- **Labware, plasticware, and glassware.** Cell culture dishes, flasks, plates, and tubes, coverslips, microscope slides, cell scrapers, cell strainers, cryovials, cuvettes, beakers, boiling flasks, dewar flasks, disposable pipettes, serological pipettes, pipette tips, extraction thimbles, film and foil wrapping, filter papers, funnels, graduated cylinders, lab coats, laboratory spatulas, latex gloves, media bottles, microcentrifuge tubes, PCR tube strips, petri dishes, spray bottles, square glass bottles, stir bars, storage jars, vials, wash bottles, weigh boats, and weighing papers.

D Empirical Bayes Shrinkage of Market-Level Price Effects

Each of the 46 treated markets m yields a difference-in-differences estimate $\hat{\beta}_m$ of the post-merger change in log prices from equation (2), estimated on the university–product market–year panel with 2013 spending weights and a standard error $\widehat{se}_m$ clustered by market. Some markets have few observations, so the raw estimates overstate the true dispersion in price effects, and the noisiest would otherwise enter the exposure measure at face value.

We therefore apply normal–normal empirical Bayes shrinkage, with sampling model $\hat{\beta}_m | \beta_m \sim N(\beta_m, \widehat{se}_m^2)$ and prior $\beta_m \sim N(\mu, \tau^2)$. We estimate $\hat{\tau}^2 = \max\{0, \text{Var}(\hat{\beta}_m) - \overline{\widehat{se}_m^2}\} = 0.031$ by method of moments and $\hat{\mu} = 0.18$ as the precision-weighted mean of the $\hat{\beta}_m$, which exceeds the unweighted mean of 0.071 because tightly estimated markets have larger price increases.

The posterior mean shrinks each estimate toward $\hat{\mu}$ in proportion to its noise,

$$\hat{\beta}_m^{EB} = w_m \hat{\beta}_m + (1 - w_m) \hat{\mu}, \quad w_m = \frac{\hat{\tau}^2}{\hat{\tau}^2 + \widehat{se}_m^2}.$$

Cross-market heterogeneity is large relative to sampling noise, with $\hat{\tau}^2 / (\hat{\tau}^2 + \overline{\widehat{se}_m^2}) = 0.94$, so shrinkage is mild. The median weight is 0.99, eight of 46 markets have $w_m < 0.9$, and the smallest weight is 0.60. The average estimate moves by 0.011 log points, the cross-market standard deviation falls from 0.182 to 0.164, and raw and shrunk estimates correlate at 0.996.

The shrunk coefficients are the market-level price effects $\hat{\beta}_m^p$ used in the rest of the paper. We apply the same procedure to the market-level spending coefficients ($\hat{\tau}^2 = 0.093$, $\hat{\mu} = 0.37$), which are used only descriptively.

E Survey of Life Sciences PIs

We survey active U.S. life sciences PIs about how they manage supply costs, recruiting respondents by email in two stages. A pilot of 54 PIs at ten universities, drawn in equal numbers from assistant, associate, and full professors, gauged response rates. The main wave, launched on August 13, 2026, targets large R1 and R2 universities. At each we select up to three life sciences departments, collect the faculty listed on each department's website with the assistance of a large language model (Claude Opus 5), and invite a random half of them. As of August 31, 2026, we have emailed 255 PIs at 14 public and private institutions and received 19 responses, a response rate of about 7 percent. The questionnaire is administered through Google Forms and was deemed exempt by the Harvard University IRB (IRB26-0945). Data collection is ongoing, so we treat the results as preliminary. The full instrument follows.

* Indicates required question

About your lab

1. What year did you start running your own lab? *

2. What best describes your field? *

Check all that apply.

- ☐ Life sciences, mainly experimental (bench work)
☐ Life sciences, mainly computational
☐ Chemistry
☐ Engineering, physics, or materials science
☐ Prefer not to say
☐ Other: _____

3. Which best describes your the institution your lab is based in? *

Mark only one oval.

- ☐ Public University
☐ Private University
☐ Academic medical center or hospital
☐ Nonprofit or independent research institute
☐ Prefer not to say
☐ Other: _____

4. How many people work in your lab, including you? Type NA if you'd rather not answer. *

How your lab buys supplies

5. Who places most supply orders for your lab? *

Mark only one oval.

- ☐ I do
- ☐ A lab manager or technician
- ☐ Students and postdocs order what they need
- ☐ A purchasing office
- ☐ Prefer not to say
- ☐ Other: _____

6. Which of the following does your lab currently do? Select all that apply. *

Check all that apply.

- ☐ Compare prices across suppliers before ordering
- ☐ Request quotes or negotiate prices with vendors
- ☐ Order through your institution's purchasing system to get negotiated prices
- ☐ Maintain a standing list of preferred products or vendors
- ☐ Track supply spending against the budget regularly
- ☐ Set rules about what lab members can order without approval
- ☐ None of these
- ☐ Prefer not to say
- ☐ Other: _____

7. Now think back to your first two years running your own lab. Which of these did you ^{*} do then? Select all that apply.

Check all that apply.

- ☐ Compare prices across suppliers before ordering
- ☐ Request quotes or negotiate prices with vendors
- ☐ Order through your institution's purchasing system to get negotiated prices
- ☐ Maintain a standing list of preferred products or vendors
- ☐ Track supply spending against the budget regularly
- ☐ Set rules about what lab members can order without approval
- ☐ None of these
- ☐ I do not remember
- ☐ Prefer not to say
- ☐ Other: _____

8. How did you learn to manage supply purchasing for a lab? Select all that apply. ^{*}

Check all that apply.

- ☐ Trial and error once I had my own lab
- ☐ From my PhD or postdoc advisor
- ☐ From a lab manager or technician
- ☐ From colleagues or other PIs
- ☐ Institutional training or onboarding
- ☐ Someone else handles it
- ☐ I have not really learned it
- ☐ Prefer not to say
- ☐ Other: _____

When costs go up

9. Think of the most recent time the price of something your lab buys went up noticeably. What did your lab do? Select all that apply. *

Check all that apply.

- ☐ Switched to a cheaper supplier or brand
- ☐ Bought in larger quantities, or coordinated orders with other labs
- ☐ Made reagents or solutions in-house instead of buying pre-made kits
- ☐ Used less material per experiment (smaller volumes, fewer replicates)
- ☐ Ran fewer experiments, or dropped conditions
- ☐ Delayed or reduced hiring (postdoc, technician, student support)
- ☐ Delayed or reduced equipment purchases
- ☐ Shifted costs onto a different grant or fund
- ☐ Applied for additional funding
- ☐ Nothing, we absorbed it
- ☐ This has not happened, or I have not noticed it happening
- ☐ Prefer not to say
- ☐ Other: _____

10. By roughly how much would the price of something you buy regularly have to rise before you changed something about how the lab works? *

Mark only one oval.

- ☐ Less than 5%
- ☐ 5 to 10%
- ☐ 10 to 25%
- ☐ 25 to 50%
- ☐ More than 50%
- ☐ I would not change anything
- ☐ Prefer not to say

11. Since you started running your lab, has a change in supply costs ever affected how you work? If so, describe what happened and what you did. Anything else you would like to add about supply costs or purchasing is welcome here. *

12. Roughly what does your lab spend per year on consumable supplies and reagents? Please exclude salaries, instruments and equipment, and core facility fees (sequencing, mass spec, imaging). *

Mark only one oval.

- ☐ Under \$10,000
- ☐ \$10,000 to \$49,999
- ☐ \$50,000 to \$149,999
- ☐ \$150,000 to \$499,999
- ☐ \$500,000 or more
- ☐ Not sure
- ☐ Prefer not to stay
- ☐ Other:

13. Roughly what percentage of your lab's total annual budget does that represent? Include salaries and all other direct costs, but exclude indirect or overhead costs. Type NA if you'd rather not answer. *

14. Thermo Fisher acquired Life Technologies (Invitrogen, Applied Biosystems, Gibco) * in early 2014. Did you notice any price change in those brands at the time?

Mark only one oval.

- ☐ Yes, noticeable increase
- ☐ Yes, slight increase
- ☐ No change I noticed
- ☐ I was not running a lab in 2014
- ☐ I did not buy those brands
- ☐ Do not remember
- ☐ Prefer not to say
- ☐ Other:

15. We are also speaking with PIs for about 30 minutes about how labs manage supply costs. If you would be willing to have that conversation, please leave your email address. This is optional, and your address will be stored separately from your responses and used only to contact you about an interview.

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